IPv6

Tecnologie e Servizi di Rete

Computer Network Technologies and Services

Internet Architecture and Technologies



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IPv6

Motivation and History



Motivation

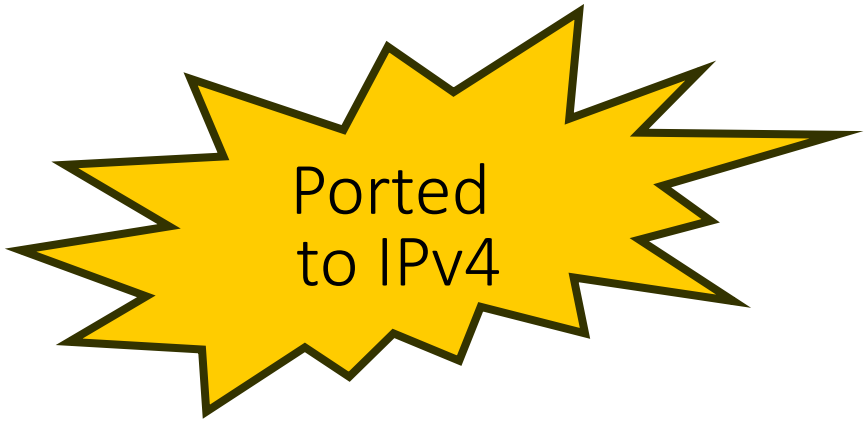
- Why a new version of the IP protocol?

- Only one true answer:

“A larger address space”

Motivation

- Additional answers:
  - More efficient on LANs
  - Multicast and Anycast
  - Security
  - Policy routing
  - Plug and Play
  - Traffic Differentiation
  - Mobility
  - Quality of Service support



The long way to IPv6 adoption

- Long time for defining IPv6 and migrating to it
- Problems needed an interim solution in IPv4
  - Port IPv6 solutions to IPv4
  - Find workaround (e.g., NAT)
- When IPv6 reached "production" stage, many IPv4 solutions were acceptable

Why are IPv4 addresses scarce?

- 32-bit long addresses → about 4 billion addresses!
- However:
  - Only part of the addresses are assigned to stations
    - Divided into Class A, B and C
    - Addresses beginning by bx111 are used for multicast and otherwise
    - Hence, "just" 3.5 billion addresses can be used!
  - They are used hierarchically
    - The prefix used in a physical network cannot be used in a different one
    - Lots of unused addresses
- Who needs them?
  - Roughly half the world populations has home access (4 billion)
  - 5.5 billion mobile phones in the world require Internet connectivity;
  - Internet of Things (IoT) will add about another 50 billion devices in this decade.

Interim solutions to the saturation of address space

- Introduction of network with "tailored" size
  - Netmask
- Private addresses
  - Intranet, RFC 1918
  - Not enough to solve the problem
    - It should be used in conjunction with NAT or ALG
- Network Address Translator (NAT)
  - Extremely popular but…
    - it breaks end-to-end reachability, increases processing load at gateways, adds complexity
  - Carrier-Grade NAT (CGN, RFC 6598, shared space 100.64.0.0/10): today the main way ISPs stretch IPv4
- ALG (Application Layer Gateway)



Has all this been effective?

Who assigns IP addresses?

- IANA assigns blocks of /8 IP addresses to each Regional Internet Registry (RIR)
  - AFRINIC – Africa
  - APNIC – East Asia, Australia and Oceania
  - ARIN – USA, Canada and some Caribbean islands
  - LACNIC – South America, Mexico and some Caribbean islands
  - RIPE NCC – Europe, Middle East and Central Asia

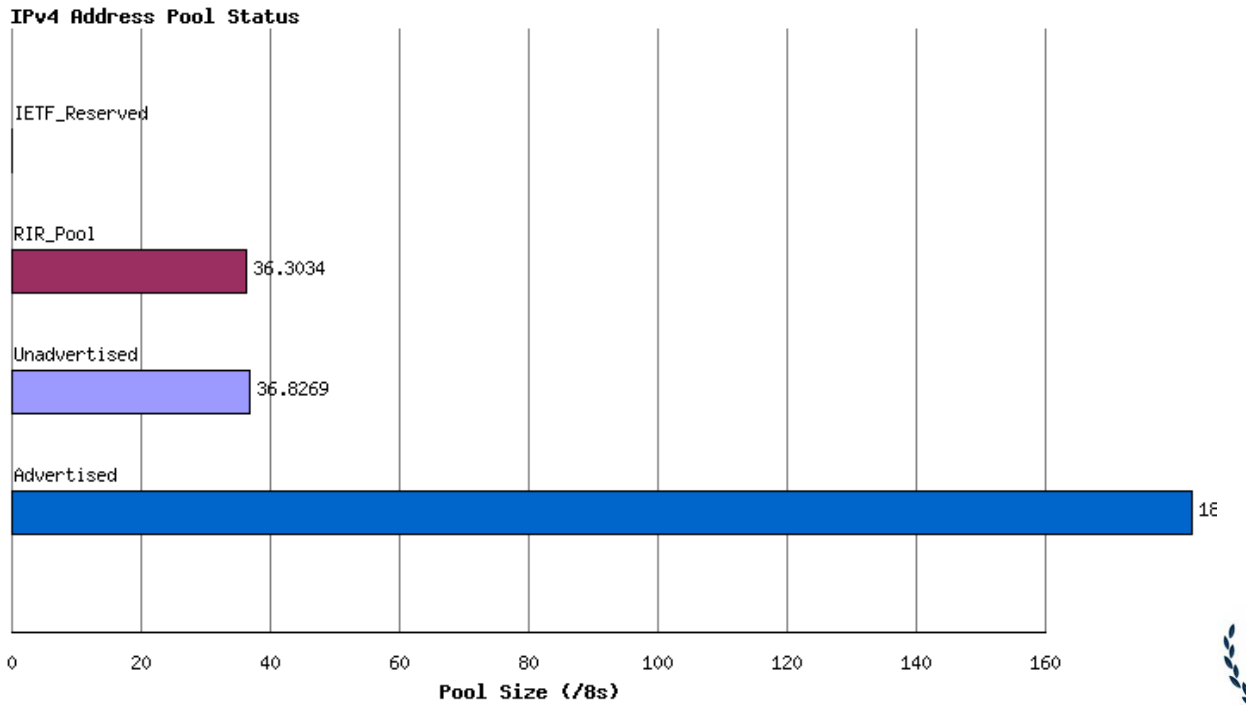


- IANA pool ran out in Feb. 2011, only RIRs have now limited /8 blocks

- Subsequently, RIRs split the blocks into smaller blocks and assign them to respective National Internet Registries (NIRs) and Local Internet Registries (LIRs)

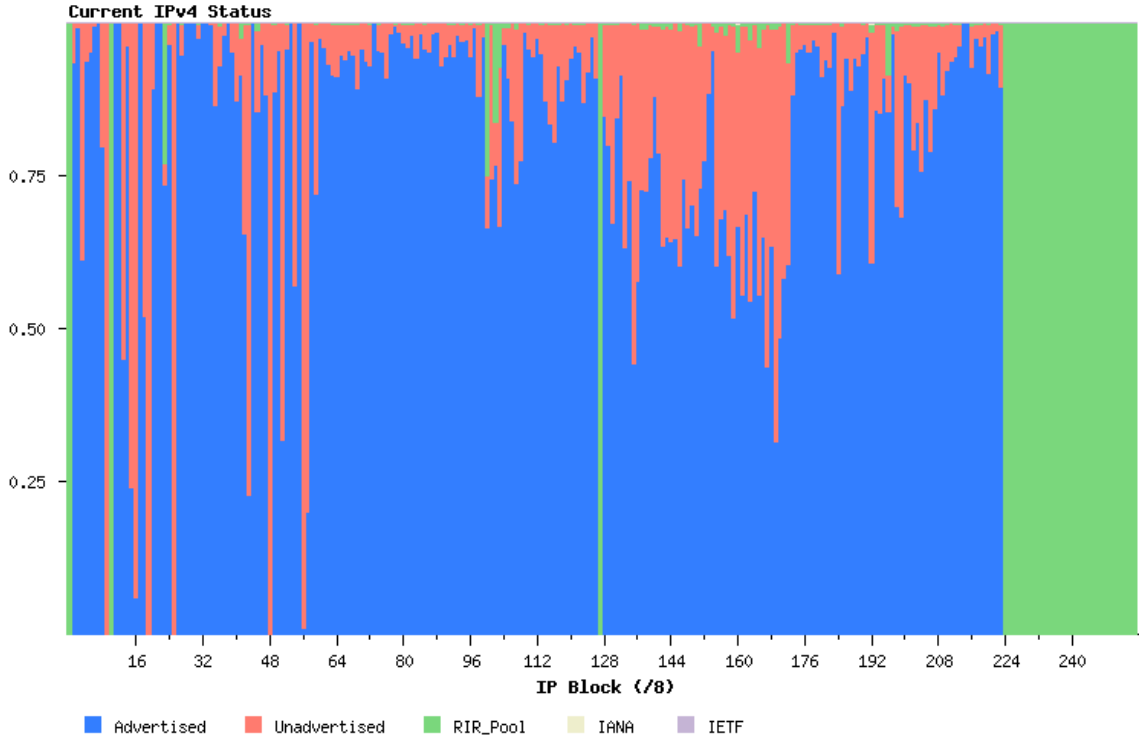
IPv4 Address Pool status

- Any individual IPv4 address can be in any one of five states:
  - reserved for special use, or
  - part of the IANA unallocated address pool,
  - part of the unassigned pool held by an RIR,
  - assigned to an end user entity but unadvertised by BGP, or
  - assigned and advertised in BGP



IPv4 Address Pool status

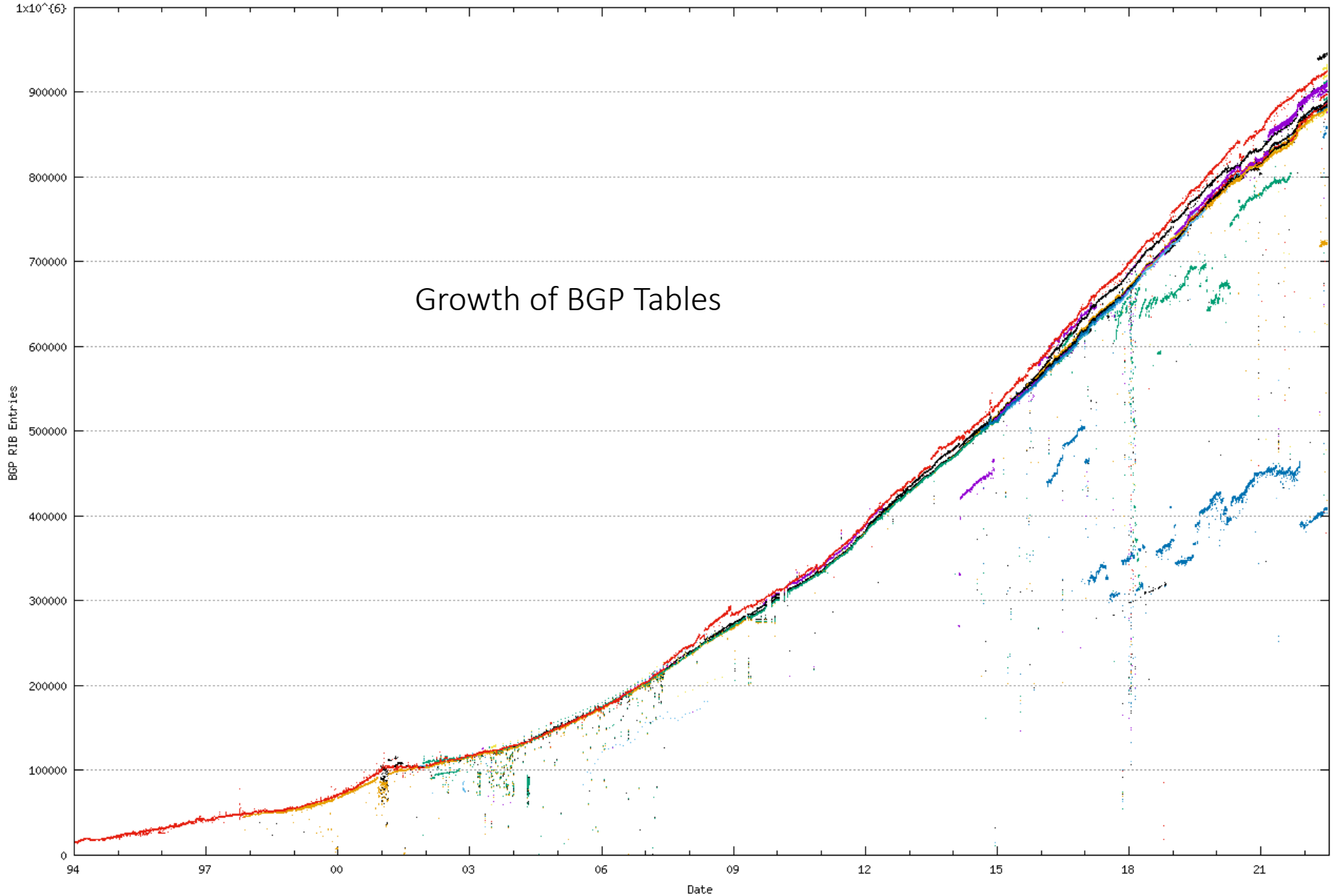
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  - reserved for special use, or
  - part of the IANA unallocated address pool,
  - part of the unassigned pool held by an RIR,
  - assigned to an end user entity but not advertised in the routing system, or
  - assigned and advertised in BGP.



Routing Scalability Issues

- Routing table size
  - Internet size
  - Each subnetwork must be advertised
- Problems
  - Router resource limitations
    - Too much information to manage
- Routing protocol limitations
  - High probability of route changes
- Mainly affecting backbone routers

Routing Scalability Issues



Possible solution with IPv4

- Aggregate multiple routes in one
  - Shorter prefix including others
  - 1.2.1.0/24, 1.2.2.0/24 …
  - 1.2.0.0/16
- CIDR (Classless Inter-Domain Routing)
- Limited by non-rational assignment of IP prefixes

Interim (IPv4) solutions to routing scalability

- CIDR
  - Classless Inter-Domain Routing
- Limiting the assignment of IP addresses
  - Regional Internet Registry: assign address blocks only to big players
  - E.g., maximum /20 (4096 addresses) network: maximum users get space from their ISP
- Scalability of routing protocols
  - No solution, at present
- Problem not completely solved
  - It is the major problem that IPv6 wanted to solve that it is still open

Birth of IPv6

- IETF Boston Meeting (1992), “Call for proposals for IPng”
  - Appointment of dedicated Working Groups
- Several proposals
  - TUBA (TCP and UDP with Bigger Addresses): adopting OSI CLNP as new IP
  - CATNIP: integration of different network (IP, CLNP, IPX) and transport (TP4, SPX, TCP, UDP) protocols
  - SIPP: incremental over IPv4
    - Fix some drawbacks
    - Minimal protocol changes: increasing the address field and eliminating unused ones
- Winning proposal: SIPP with 128-bit addresses
- RFC 1752 (1994) recommends SIPP; first IPv6 spec RFC 1883 (1995); current Internet Standard RFC 8200 (2017)

IPv6 deployment today (2026)

- Share of users reaching Google over IPv6 passed 50% for the first time in April 2026
- BGP table, Jan. 2026: ≈242,000 IPv6 prefixes vs ≈1,050,000 IPv4 prefixes
- Adoption is very uneven across countries and operators
  - Mobile networks often lead: IPv6-only access, with NAT64/464XLAT to reach IPv4 destinations
- IPv4 keeps working thanks to CGN and the address transfer market
  - Dual stack is still the norm; IPv4 cannot be switched off yet
- Live statistics: google.com/ipv6, stats.labs.apnic.net/ipv6

IPv6

Addresses



How many addresses for IPv6?

- Scientific approach followed:
  - What is the number of “usable” addresses, accounting for inefficiencies due to hierarchical allocatio
- Addressing efficiency (RFC 1715, later updated by RFC 3194):

$$HD = \frac{\log_{10}(\text{number of usable addresses})}{\log10(2\text{^number of bits})}$$

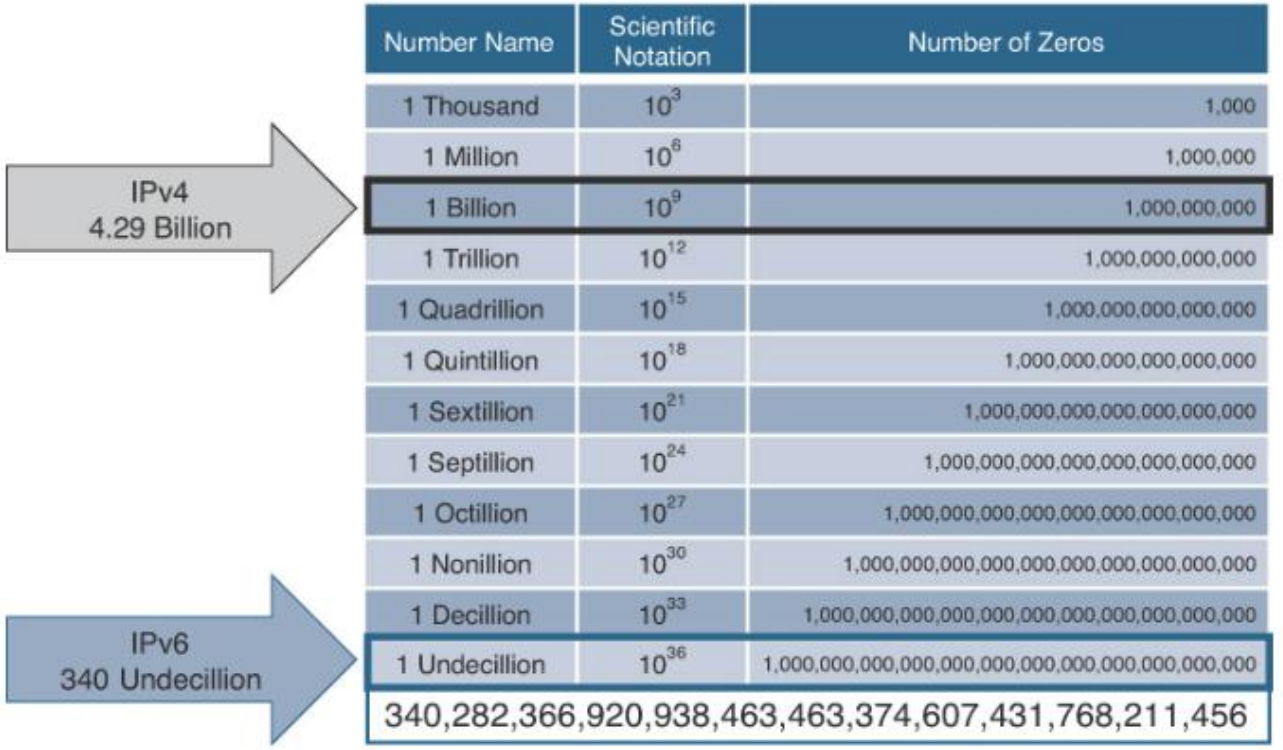
- Real-world hierarchical allocation rarely achieves an HD higher than 0.86, and values around 0.8 were *typical* or even *optimistic*. HD = 0.75 was considered *pessimistic*

64 bits	usable addresses = 2.6 E+15 (optimistic)	2.8 E+14 (pessimistic)
128 bits	usable addresses = 6.6 E+30 (optimistic)	7.9 E+28 (pessimistic)

Final decision: 128 bits

- Assuming one million billion networked stations
  - 107 bits in the minimum efficiency case, HD ≈ 0.39 (with 64 bits: HD ≈ 0.78, at the limit)

- To put it in perspective… 655.570.793.348.866.943.898.599 IPv6 addresses per m^2 of Earth surface



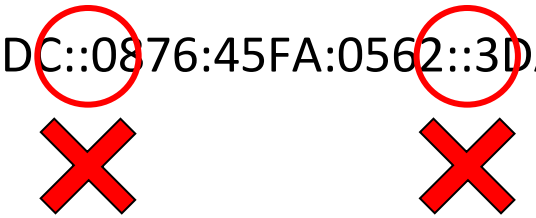
Number Name	Scientific Notation	Number of Zeros
1 Thousand	10^3	1,000
1 Million	10^6	1,000,000
1 Billion	10^9	1,000,000,000
1 Trillion	10^{12}	1,000,000,000,000
1 Quadrillion	10^{15}	1,000,000,000,000,000
1 Quintillion	10^{18}	1,000,000,000,000,000,000
1 Sextillion	10^{21}	1,000,000,000,000,000,000,000
1 Septillion	10^{24}	1,000,000,000,000,000,000,000,000
1 Octillion	10^{27}	1,000,000,000,000,000,000,000,000,000
1 Nonillion	10^{30}	1,000,000,000,000,000,000,000,000,000,000
1 Decillion	10^{33}	1,000,000,000,000,000,000,000,000,000,000,000
1 Undecillion	10^{36}	1,000,000,000,000,000,000,000,000,000,000,000,000
340,282,366,920,938,463,463,374,607,431,768,211,456		

Notation

- 8 hexadecimal numbers separated by ":" **x:x:x:x:x:x:x:x**
  - Groups of 2 bytes

- Examples:
  - FEDC:0000:0876:45FA:0562:0000:3DAF:BB01
  - 1080:0000:0000:0007:0200:A00C:3423:A089

- Shortcut rule #1: Leading 0s in each digit group can be omitted
  - 1080:0:0:7:200:A00C:3423:A089
- Shortcut rule #2: All-0s Hextets can be substituted by "::" (only once)
  - 1080::7:200:A00C:3423:A089
  - FEDC:0000:0876:45FA:0562:0000:3DAF:BB01 → FEDC::0876:45FA:0562::3DAF:BB01



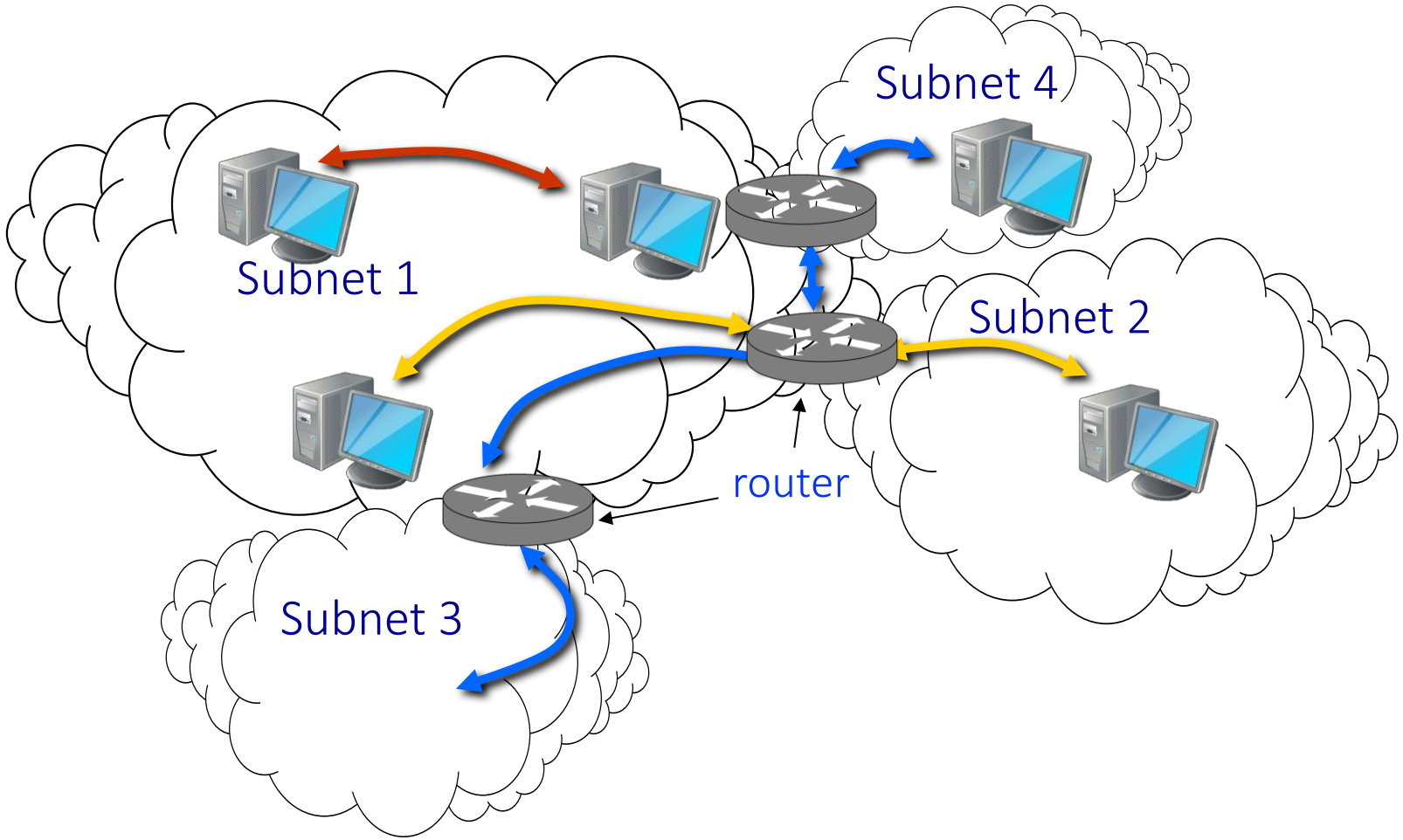
IPv6



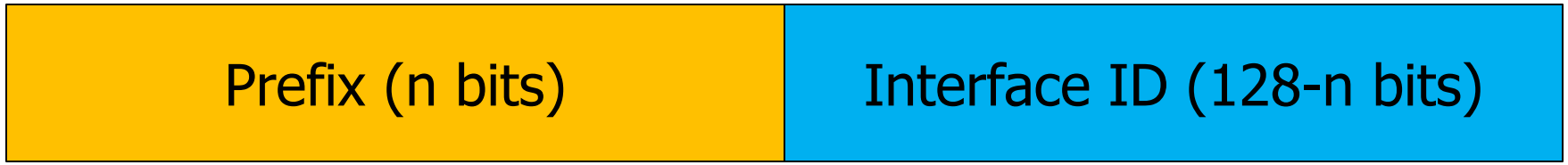
Routing and Addressing Principles

Host Addresses

Same routing principles as IPv4



Address Structure



n=64

Prefix

- Address/netmask pair is substituted by a “Prefix”
- Address/N, where N is the prefix length [bit]
- Examples:
  - Address: FEDC:0123:8700::3/48
  - Prefix: 1111111011011100000000100100111000011100000000 (also called *site prefix*)
  - Interface ID: 0000000000000000000000000000000000000000000000000000000000000011
- No address classes!

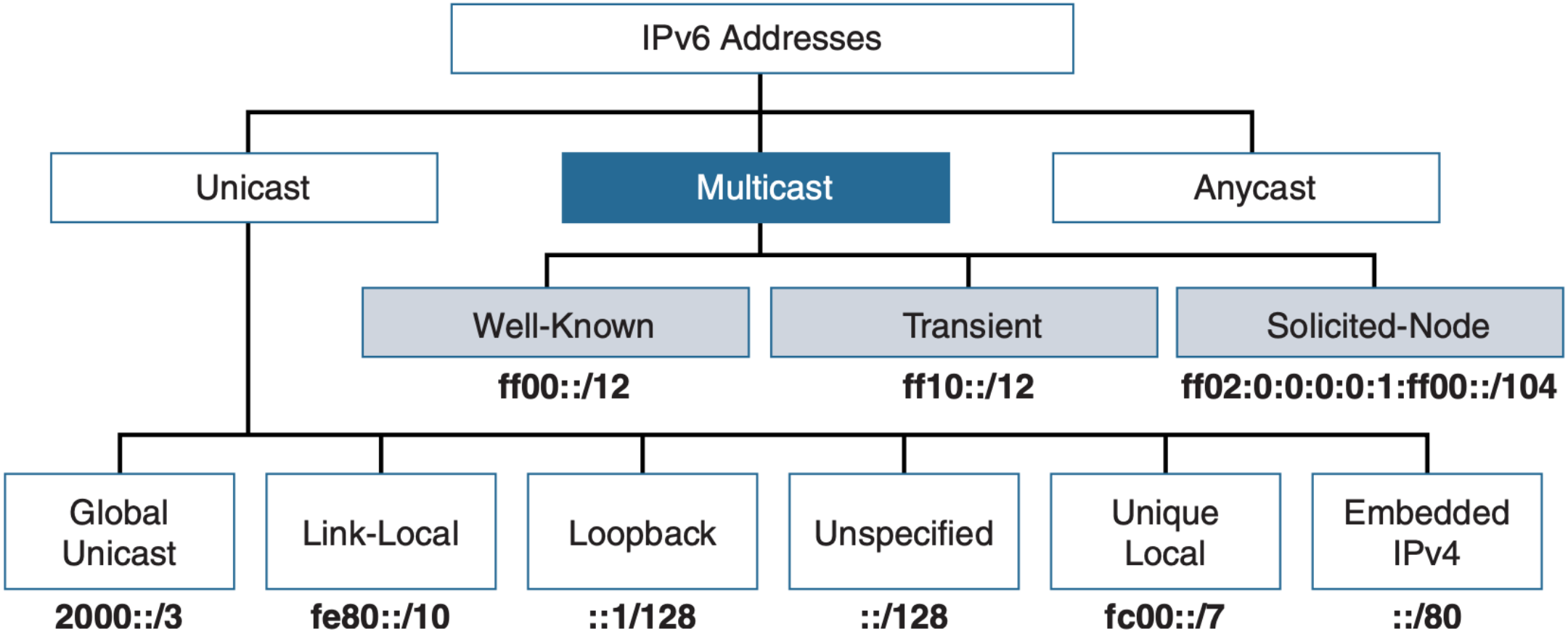
Address assignment principles

- Same as IPv4 (different terminology)
- Subnetwork: set of hosts with same prefix
- Link: physical network
- Subnetwork = Link!

- On-link hosts have the same prefix
  - Direct communication

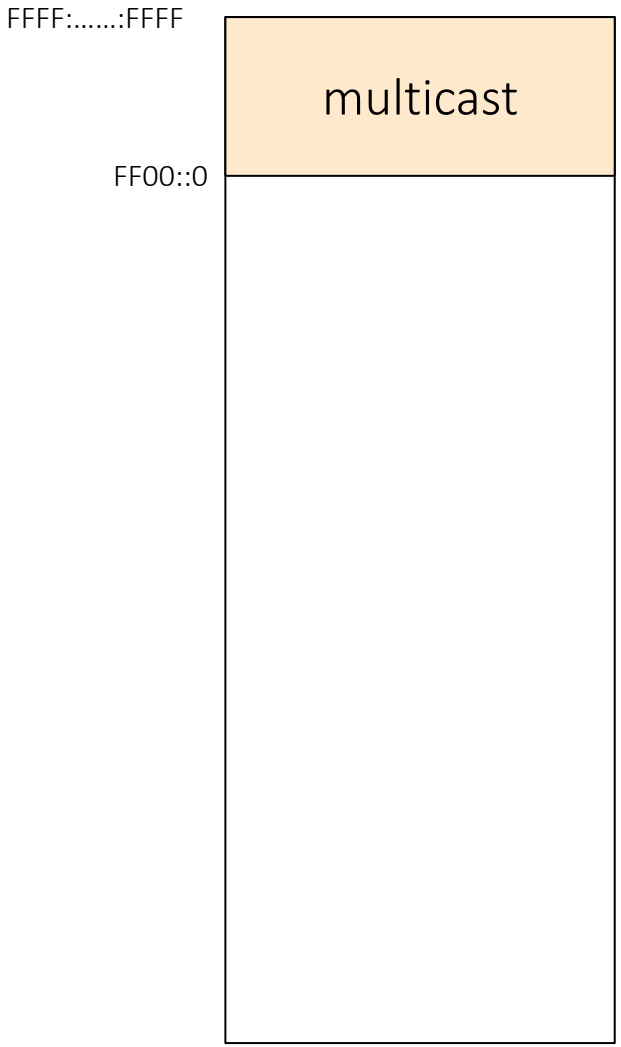
- Off-link hosts have different prefix
  - Communication through a router

IPv6 Address Space - Multicast

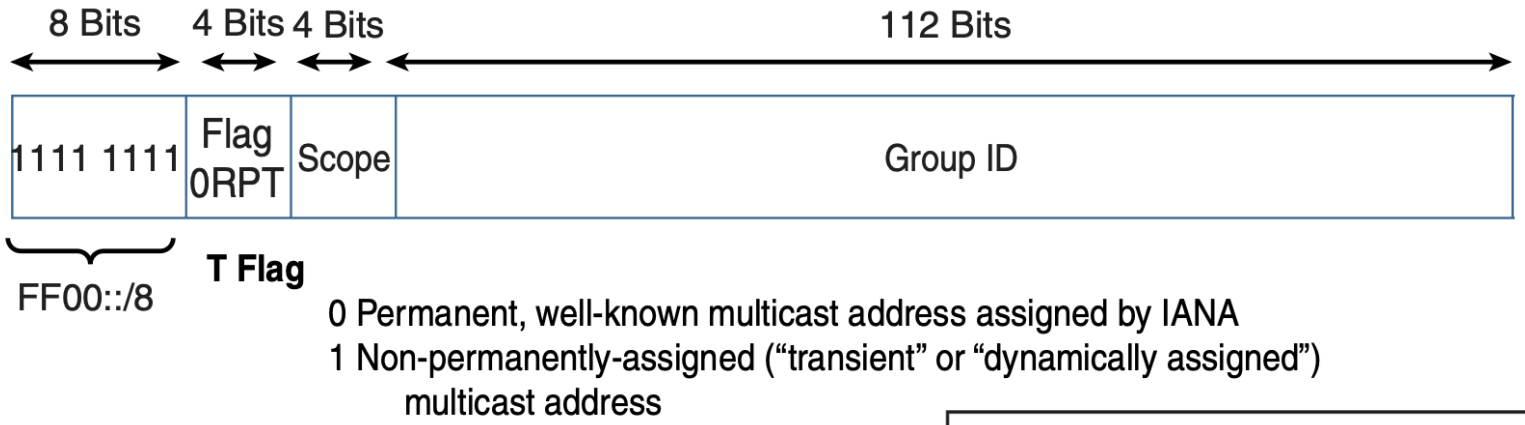


Multicast addresses

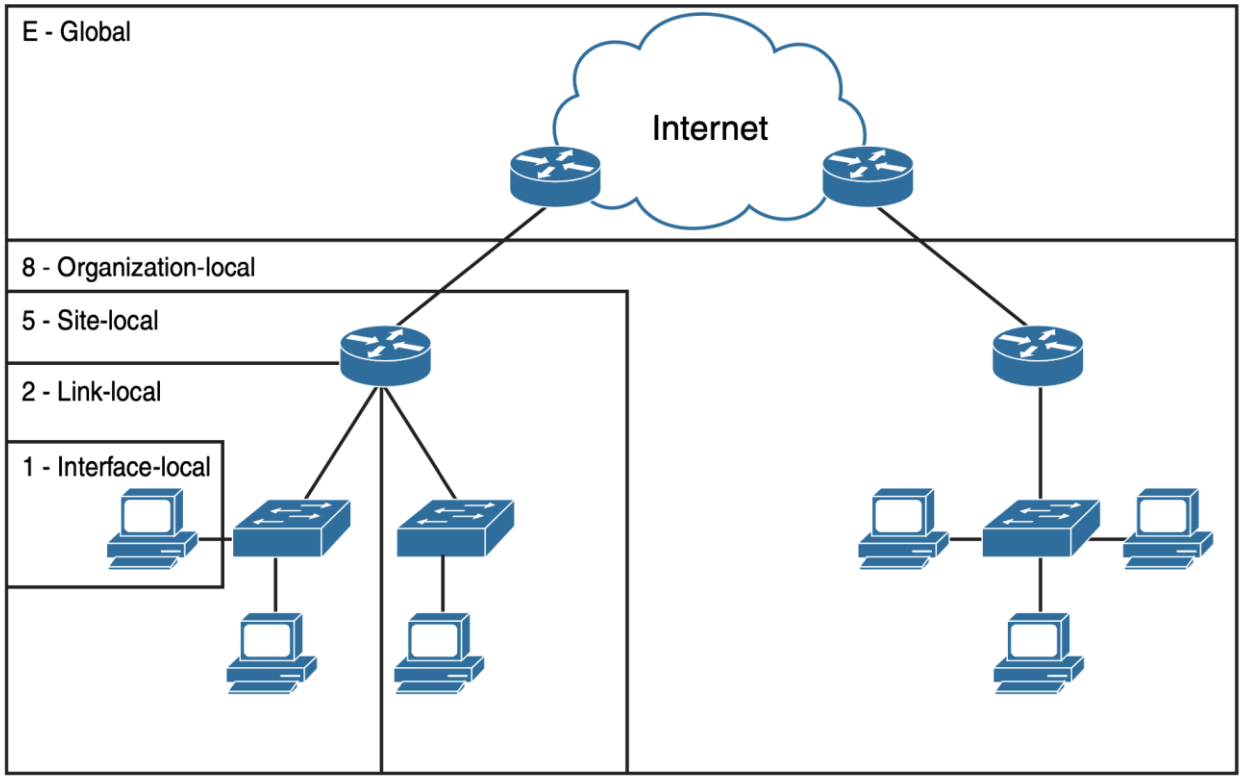
- Multicast
  - 1111 1111 ….
  - FF00::/8 → FFxx:xxx….
- equivalent to the IPv4 multicast address 224.0.0.0/4
- Further divided into:
  - *Well-known Multicast* (FF00::/12): predefined or reserved multicast addresses for assigned groups of devices (e.g., routers)
  - *Transient* (FF10::/12): dynamically assigned by multicast apps
  - *Solicited-node Multicast* (FF02:0:0:0:0:1:FF00::/104): similar to IP broadcast address in ARP (see later)



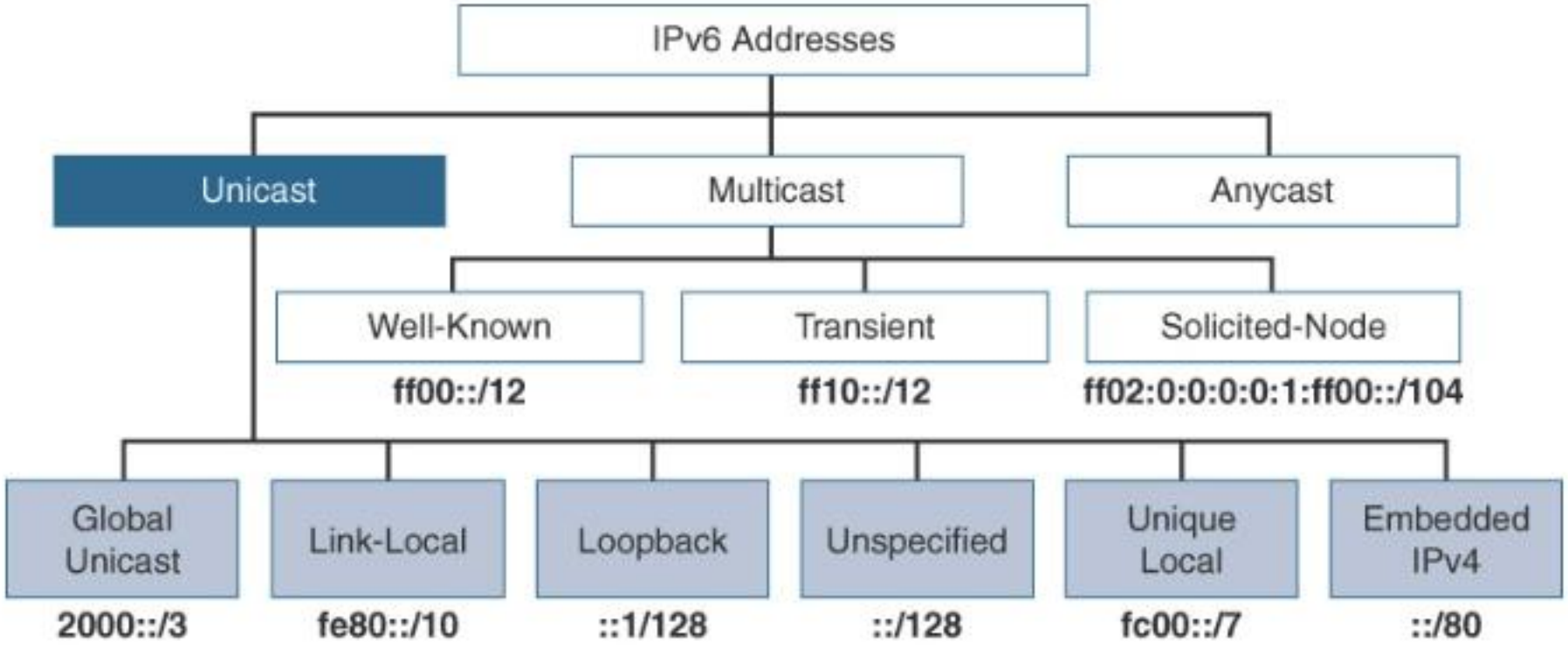
Multicast addresses



- Scope: allows devices to define the range of the multicast packet
  - how broadly to propagate the packet

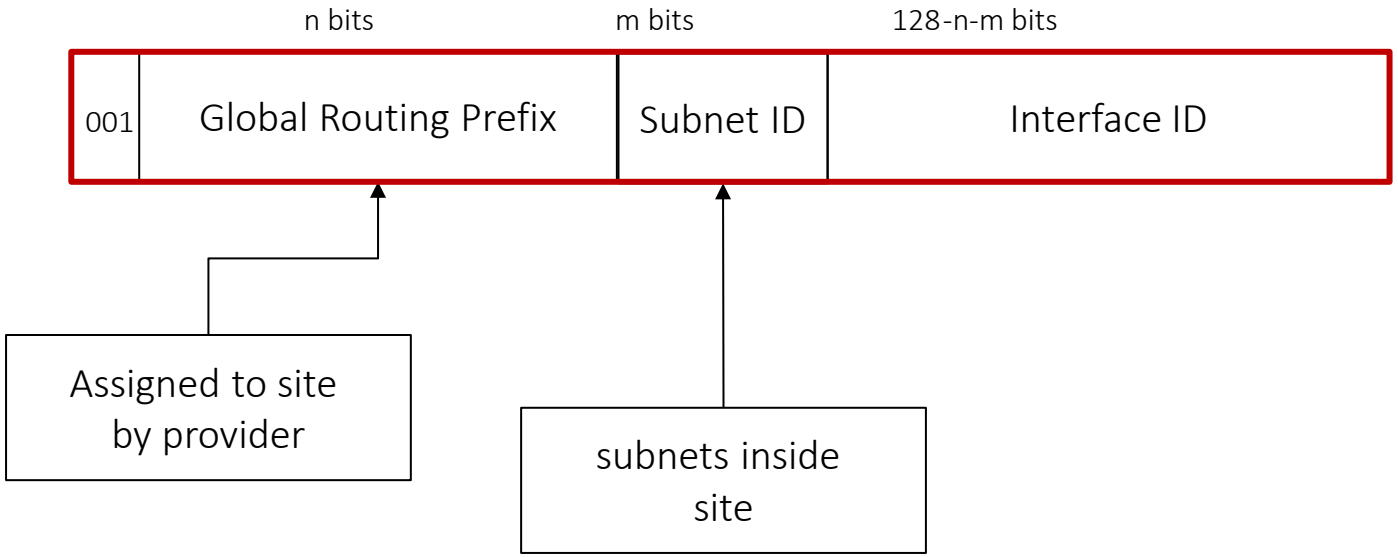


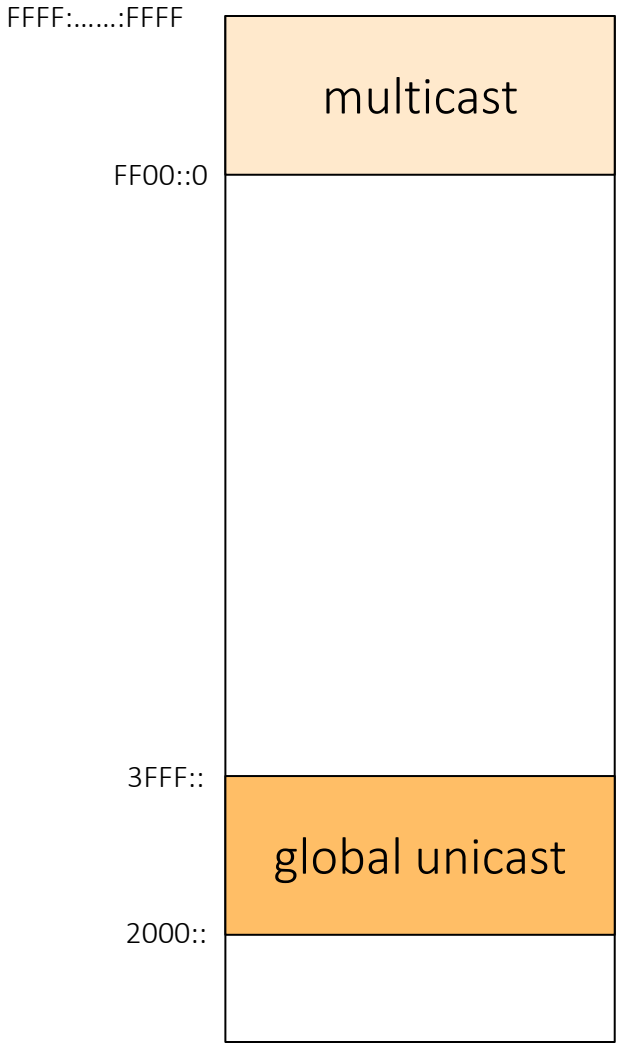
IPv6 Address Space - Unicast



Global Unicast Addresses

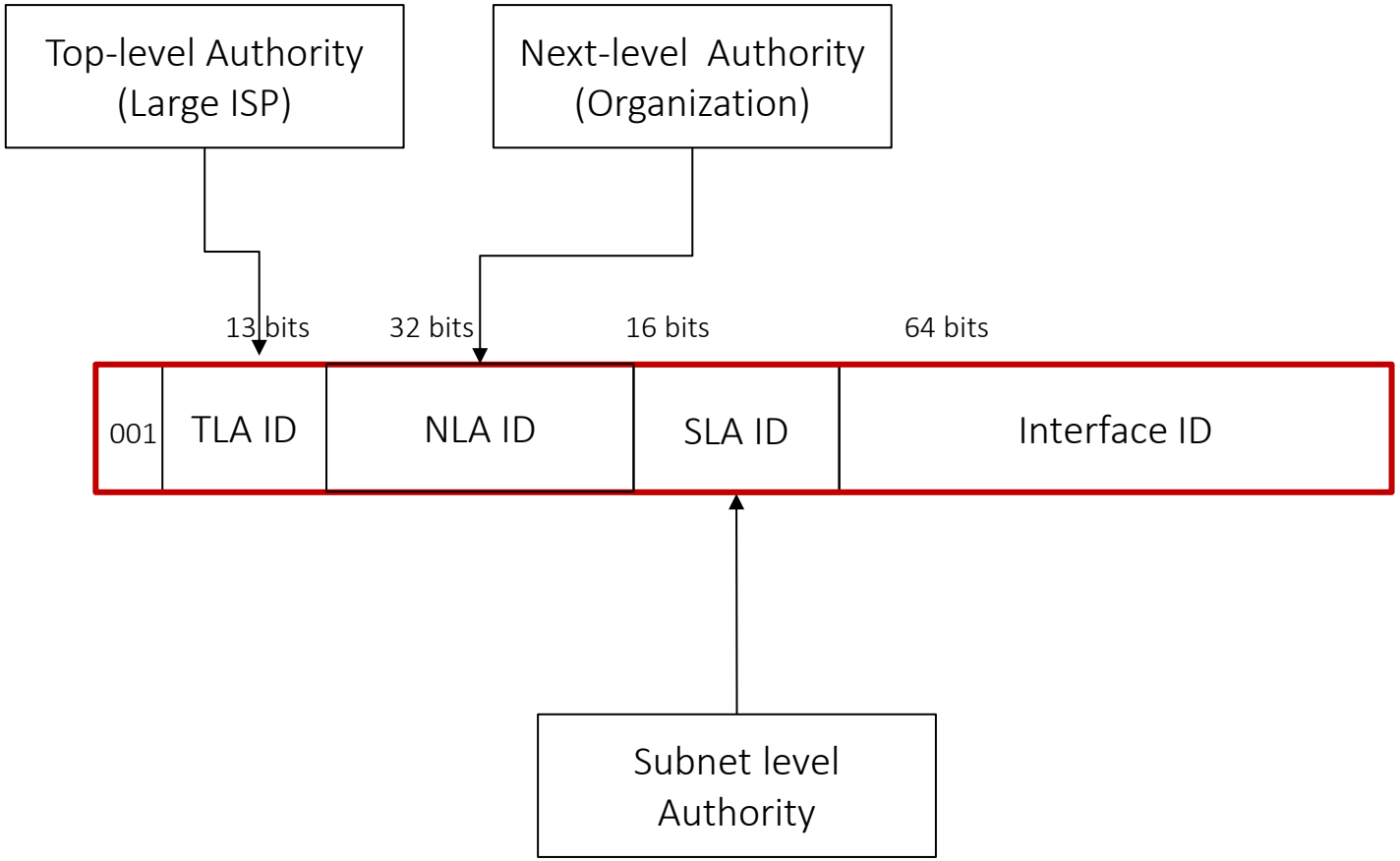
- also known as aggregatable global unicast addresses
- globally routable and reachable
- Plug and play
- equivalent to public IPv4 addresses
- Topology-based assignment with effective aggregation

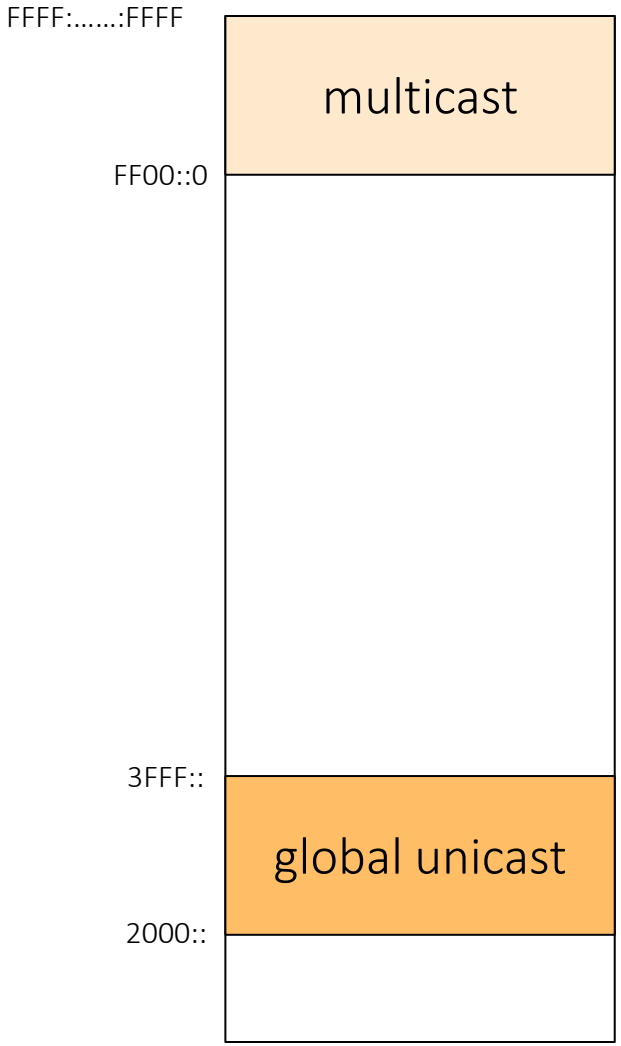




Global Unicast Addresses

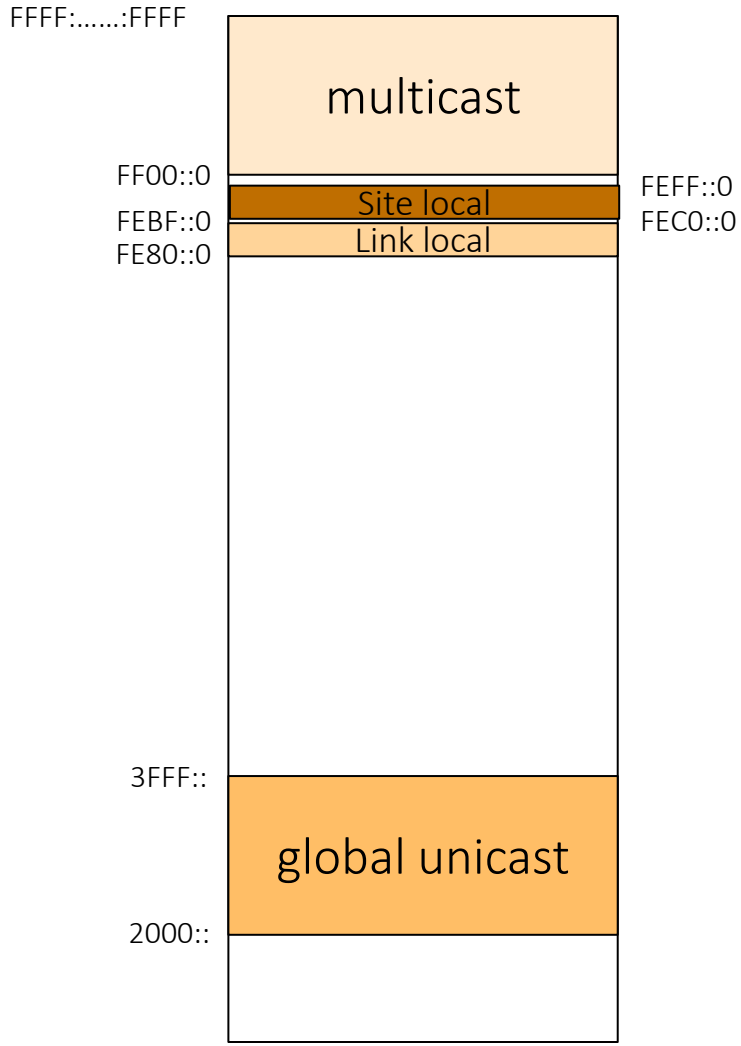
- Today's Global Routing prefix was *formerly* assigned by multi-level authorities:





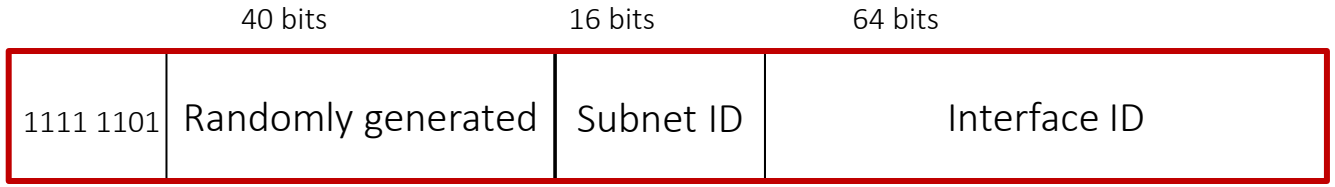
Link local/site local addresses

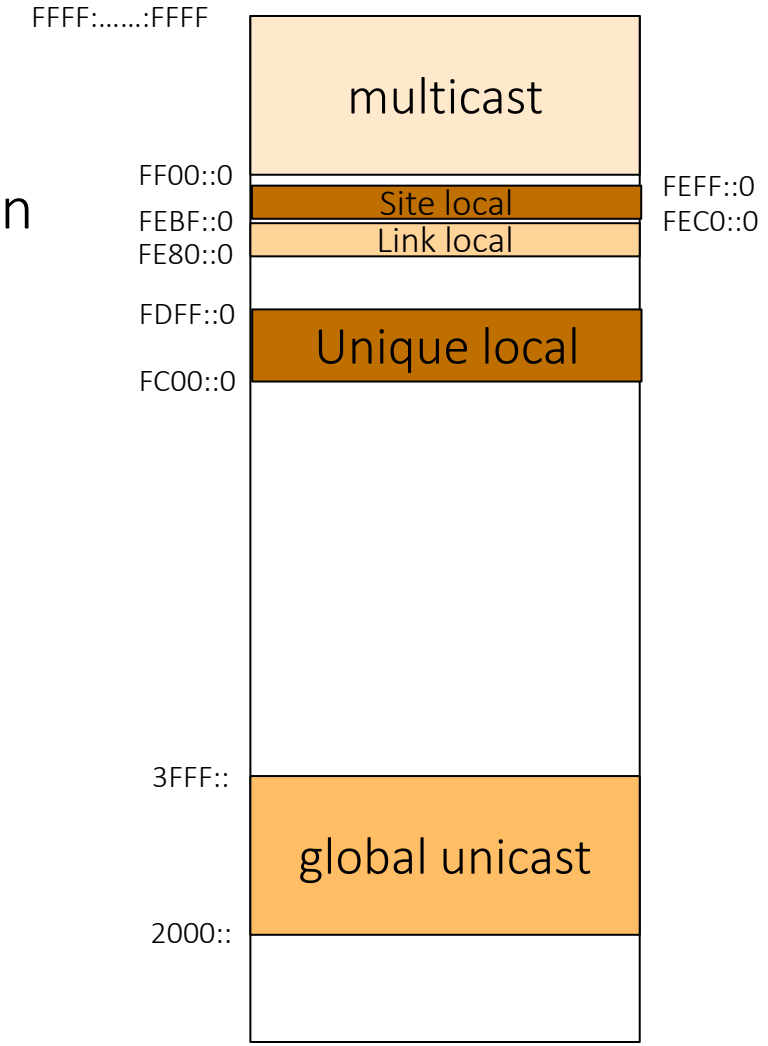
- Link/site local
  - 1111 1110 1….
- Link local
  - Used for addresses on a single link for auto-address configuration, neighbor discovery, or when no routers are present
  - Packets sent to these addresses are not supposed to be forwarded across local links
  - 1111 1110 10….
  - FE80::/64
- Site local
  - Deprecated
  - FEC0::/10



Unique Local Addresses

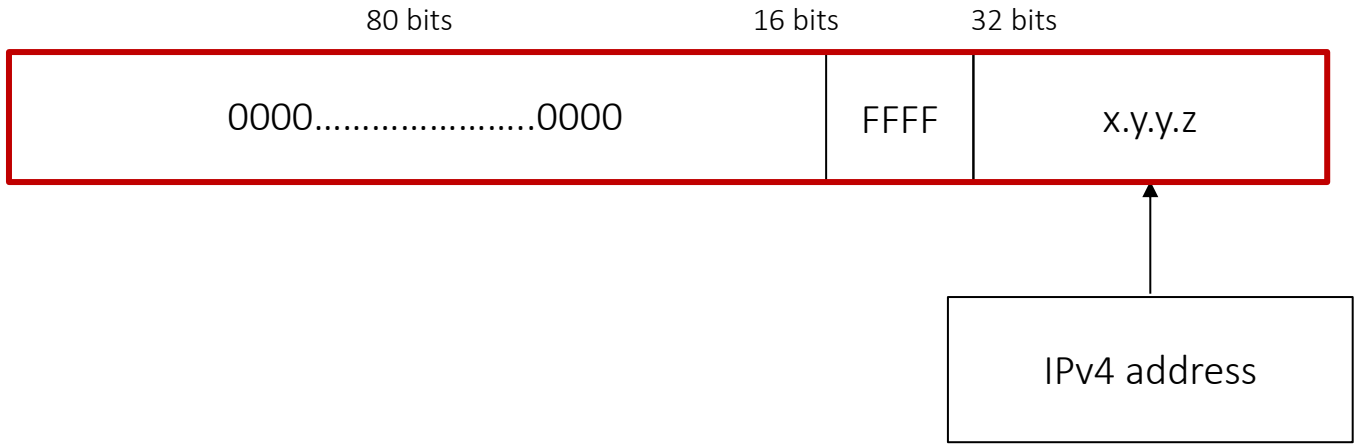
- ULA addresses can be used similarly to global unicast addresses but are for private use and should not be routed in the global Internet
- FC00::/7
- Used for devices that never need access to the Internet and never need to be accessible from the Internet
- The 8th bit is the “Local (L) Flag”, dividing the range in:
  - fc00::/8 (1111 1100): If L flag is 0, may be defined in the future
  - fd00::/8 (1111 1101): If L flag is 1, the address is locally assigned
  - fd00::/8 are the only valid ULA addresses today!

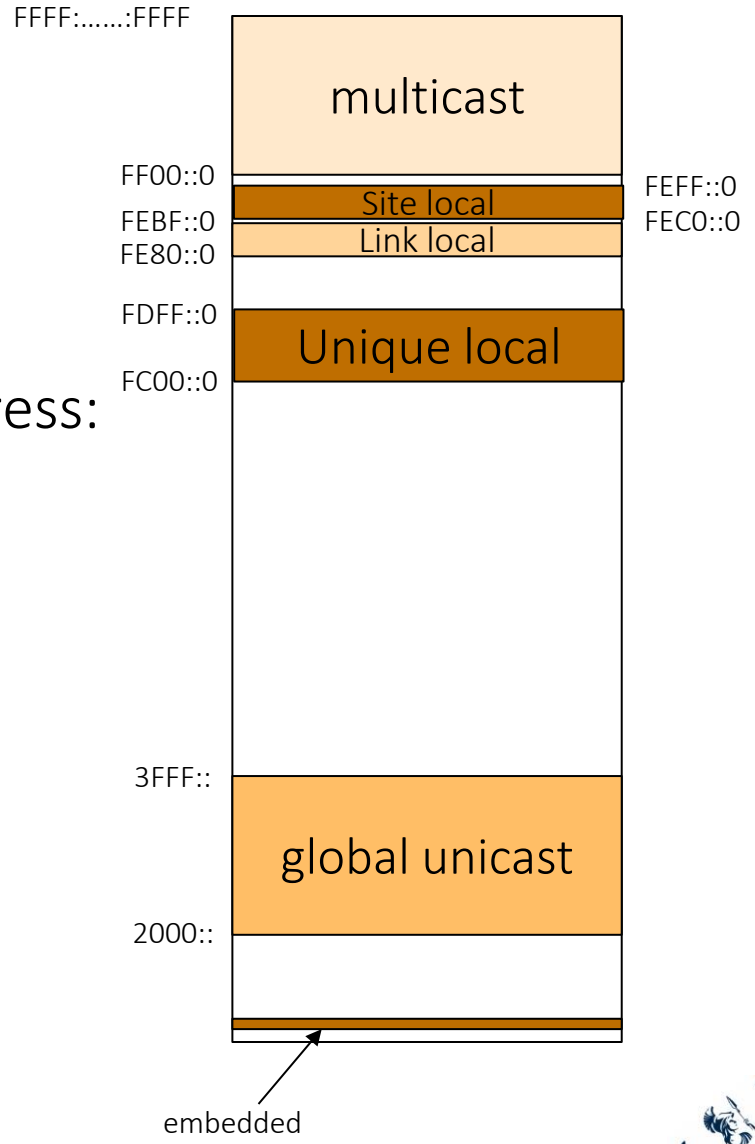




IPv4 Embedded Addresses

- used to represent an IPv4 address inside an IPv6 address
- aid the transition from IPv4 to IPv6
- IPv4 address carried in the low-order 32 bits
- e.g., IPv6 compressed format of IPv4-mapped 20.10.4.3 address:
  - ::ffff:20.10.4.3





Loopback Addresses

- Used by a node to send an IPv6 packet to itself for testing reasons
- ::1 (000…1)
- Equivalent to IPv4 address: 127.0.0.1
- Same rules as IPv4:
  - cannot be assigned to a physical interface.
  - packets with a loopback address should never be sent beyond the device.
  - routers can never forward packets with a destination address that is a loopback address.
  - device must drop a packet received on an interface if the destination address is a loopback address.

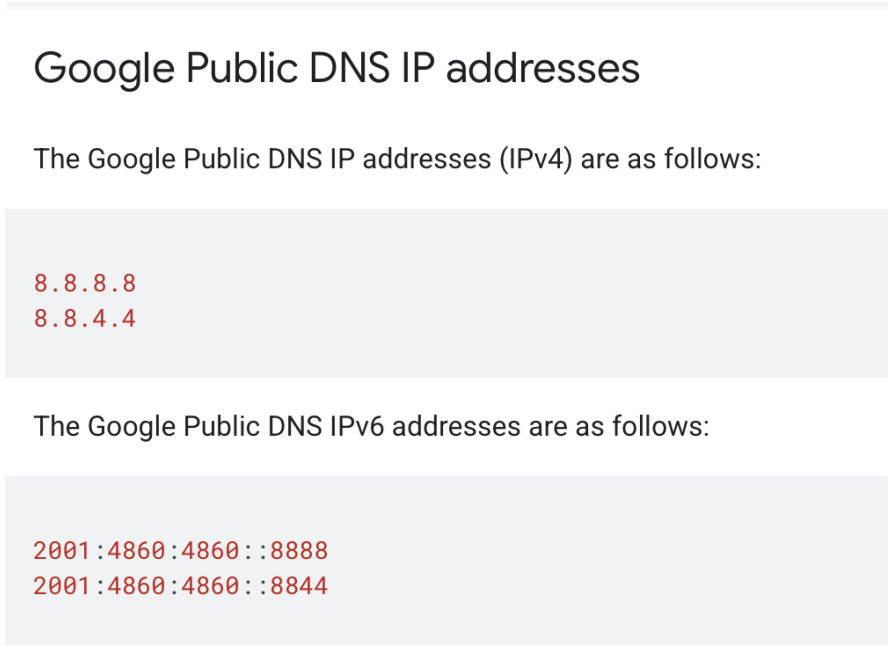
Unspecified Addresses

- An unspecified unicast address is an all-0s address
- An unspecified unicast address is used as a source address to indicate the absence of an address.
- It cannot be assigned to an interface.
- Used in duplicate address detection in ICMPv6



Anycast address

- Address that can be assigned to more than one interface (typically different devices).
  - multiple devices can have the same anycast address
- A packet sent to an anycast address is routed to the “nearest” interface having that address, according to the router’s routing table
- Anycast addresses use Global Unicast Addresses, with routers advertising the same prefix from multiple sites and BGP (global routing protocol) handling duplicates
- Now used extensively for public DNS and CDNs (Content Distribution Networks)



Google Public DNS IP addresses

The Google Public DNS IP addresses (IPv4) are as follows:

```
8.8.8.8
8.8.4.4
```

The Google Public DNS IPv6 addresses are as follows:

```
2001:4860:4860::8888
2001:4860:4860::8844
```

IPv6

Modified Protocols



What changes in the protocol architecture?

- IP
- ICMP
- ARP
  - Integrated in ICMP
- IGMP
  - Integrated in ICMP

Upgraded but not changed

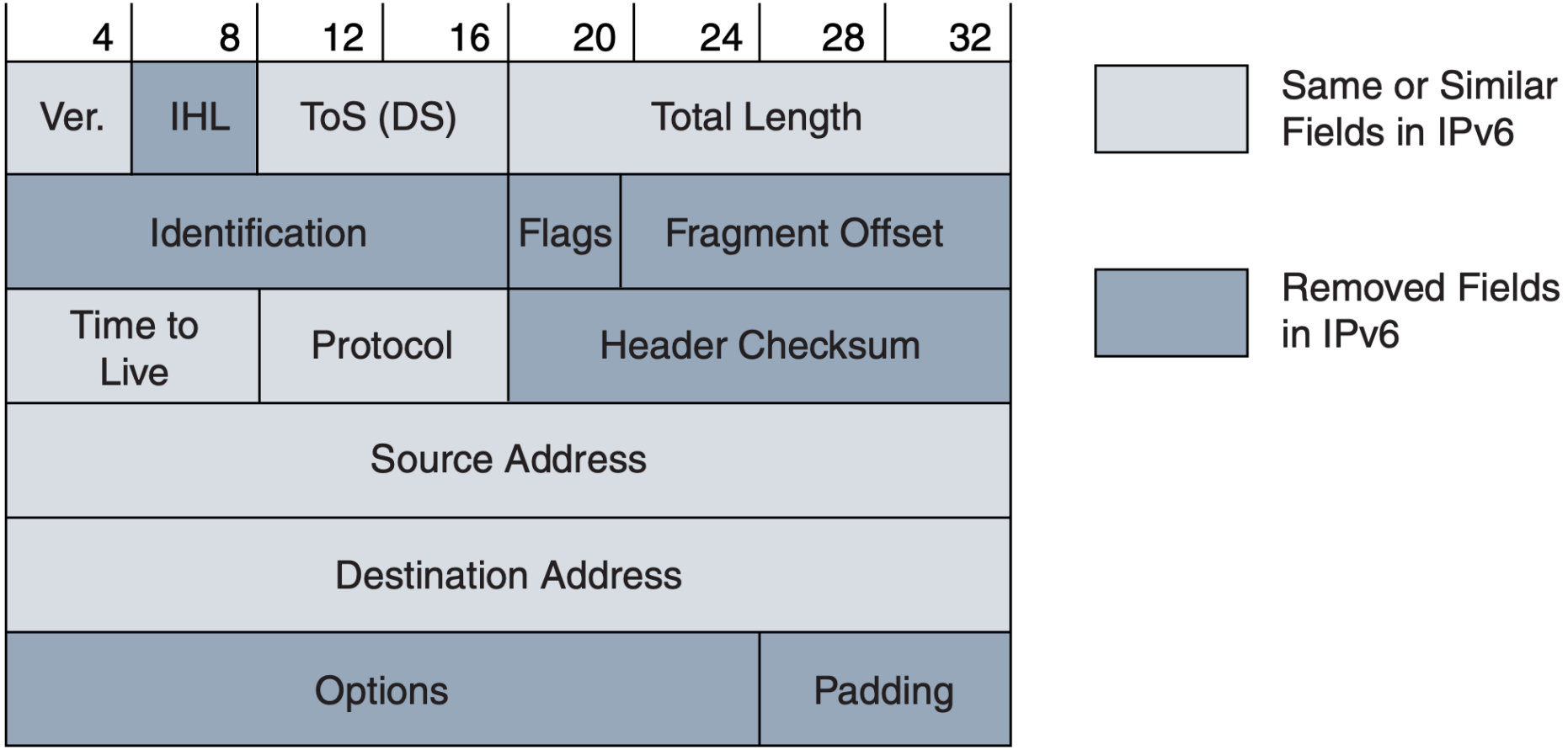
- DNS (type AAAA record)
- RIP and OSPF
- BGP and IDRP
- TCP and UDP
- Socket interface



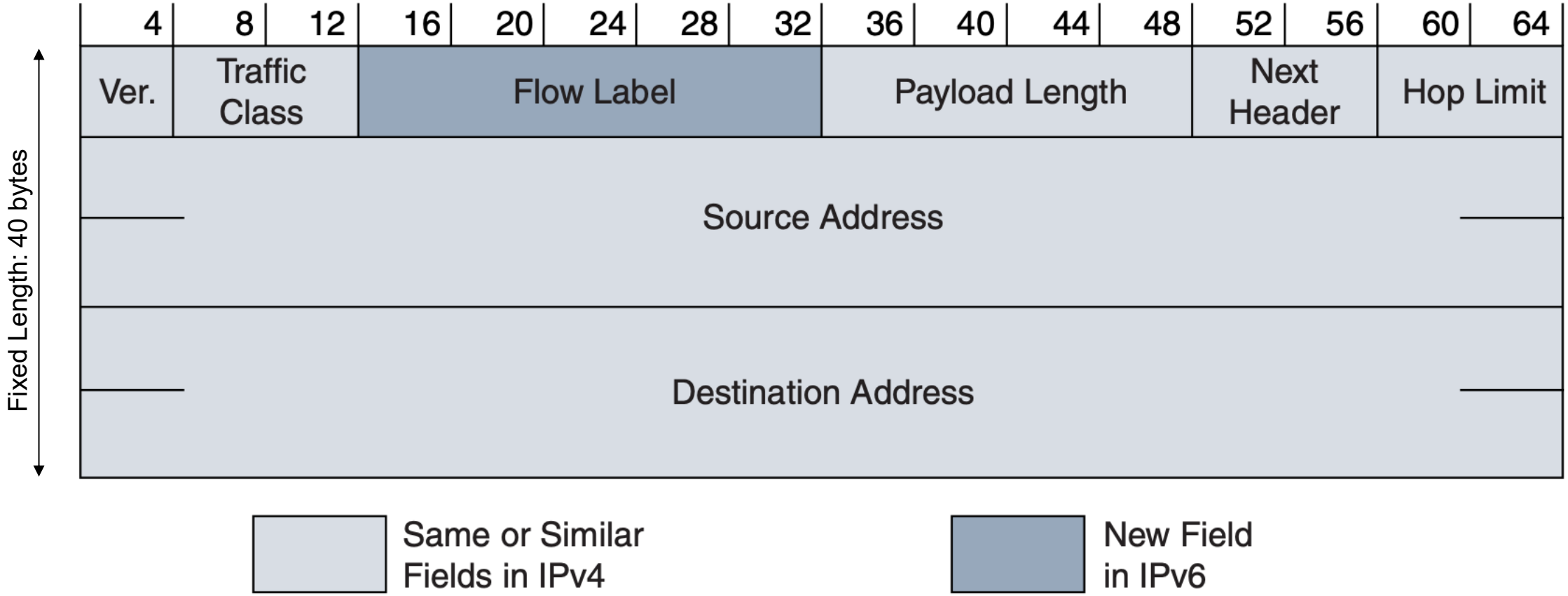
Packet Header Format



Recall the IPv4 header



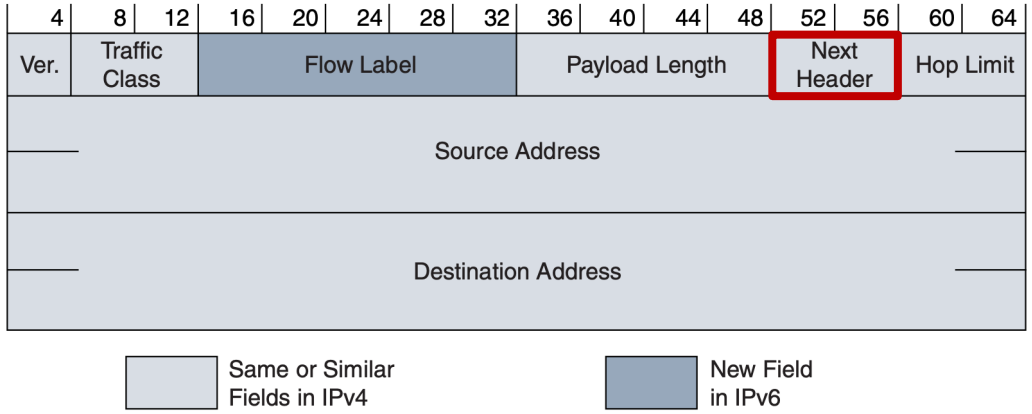
IPv6 Header


4
8
12
16
20
24
28
32
36
40
44
48
52
56
60
64
Ver.
Traffic Class
Flow Label
Payload Length
Next Header
Hop Limit
Source Address
Destination Address
Fixed Length: 40 bytes
Same or Similar Fields in IPv4
New Field in IPv6

Removed fields

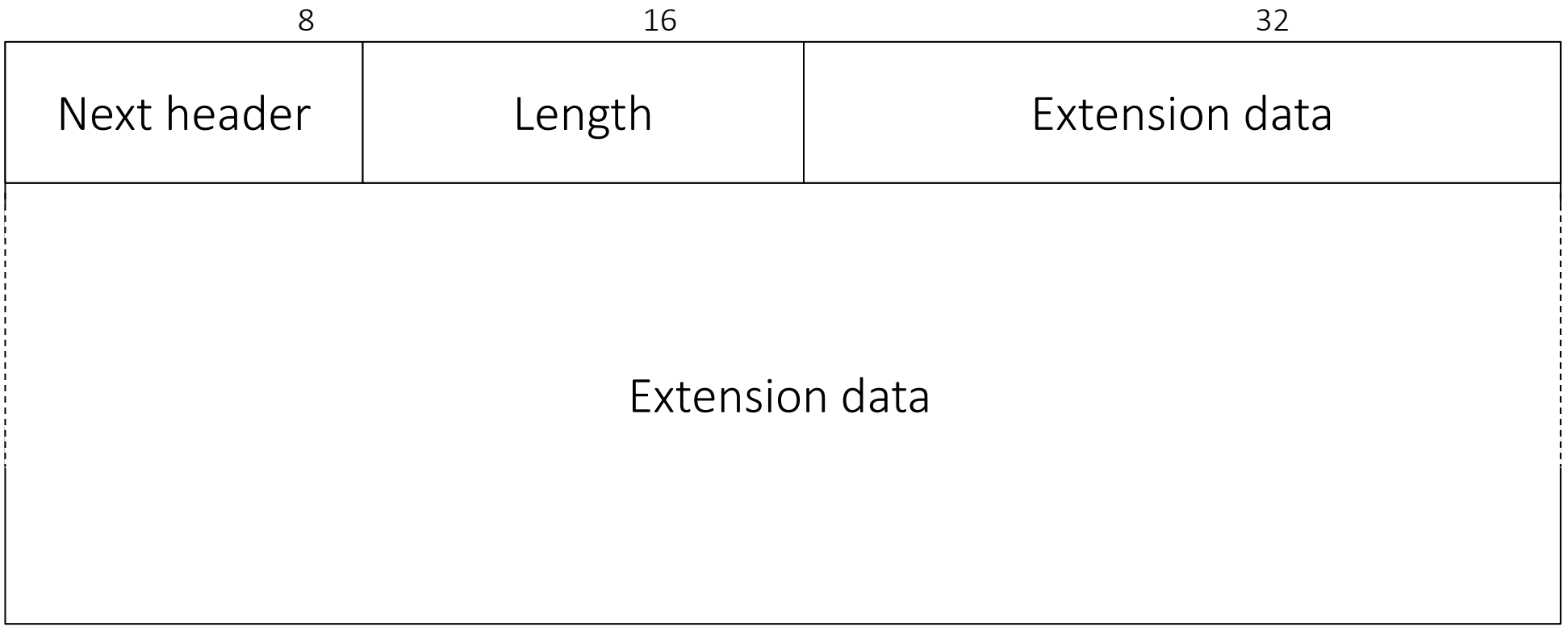
- Header Checksum
  - Redundant: Layer 2 data link technologies perform own checksum and error control.
  - Upper-layer protocols such as TCP and UDP have their own checksums
  - Note: UDP checksum becomes *mandatory* in IPv6
- Fragmentation
  - IPv6 routers do not fragment a packet unless they are the source of the packet
  - Packets larger than MTU are dropped and an ICMPv6 Packet Too Big message is returned to source
- Header Length
  - Fixed to 40 bytes

Extension headers

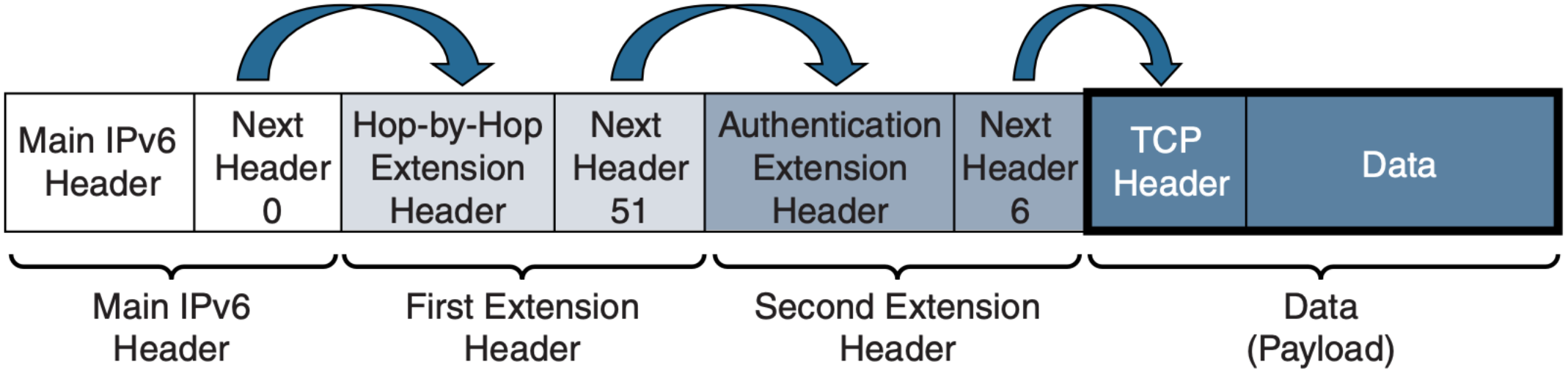


- Next Header: similar to IPv4 “Protocol” field
  - It specifies the protocol carried in the data portion of the IPv6 packet
- Additional “Next Header” fields are added in header extensions to allow chaining
- Protocols coded using same values as IPv4 (plus some)
  - 0: Hop by Hop option 6: TCP 17: UDP 43: Routing extension 44: Fragment header 50: Encapsulating Security Payload 51: Authentication 60: Destination option ……
  - 59: no next header

Extension Header Format

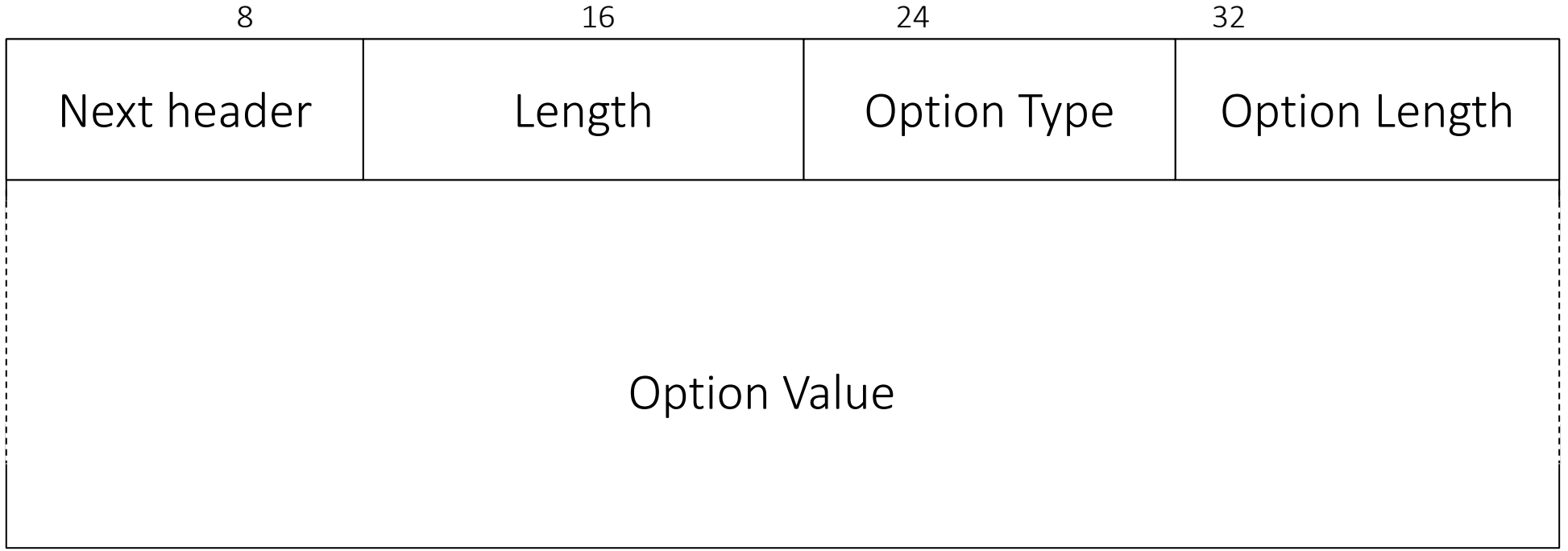


Header Chaining

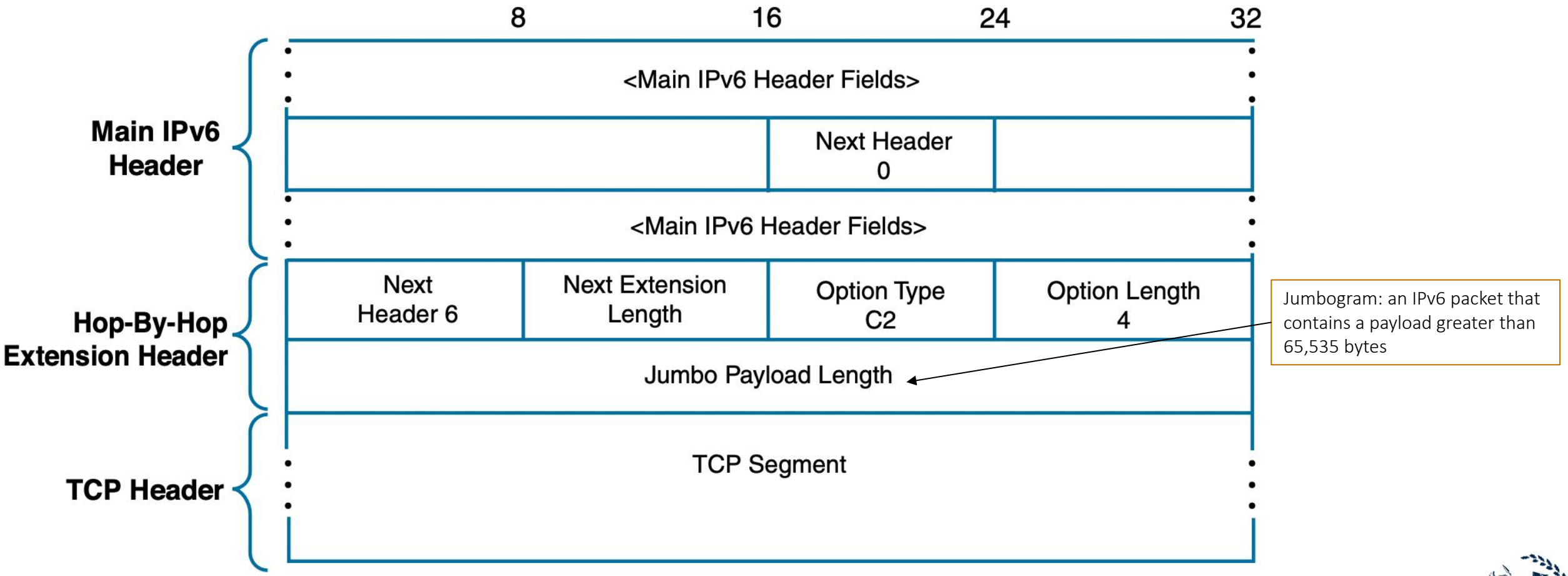


Hop-by-Hop Extension Header

- Used to carry optional information that should be examined by every router along the path of the packet
- Each option contains a set of Options Type, Options Length, and Options Value fields (TLV triplets)
- If present, it immediately follows the IPv6 header



Hop-by-Hop Extension Header - Example



More extension headers…

- Routing Extension Header
  - source routing
- Fragment Extension Header
  - used when the source needs to divide the packet into fragments, each with its own main IPv6 header and a Fragment extension header
  - the recipient of the packet reassembles the fragments,
  - Recall: unlike in IPv4, an IPv6 router does not fragment a packet unless it is the source of the packet
- Authentication and Encapsulating Security Payload Extension Headers
  - used by IPsec, a suite of protocols for securing delivery of packets in IP networks
  - the Authentication Header (AH) is used to guarantee the authenticity and integrity of a packet
  - the Encapsulating Security Payload (ESP) provides authentication, integrity, and encryption.

Interfacing with Lower Layer



Encapsulation

- IPv6 packets are encapsulated in layer-2 frames
  - For Ethernet, Type: 86DD
- They are treated as a new protocol
  - Enables dual stack (IPv4/IPv6) approach
  - Keeps running IPv4 as-is

Address Mapping

- How is the destination MAC address set when an IPv6 packet is encapsulated?

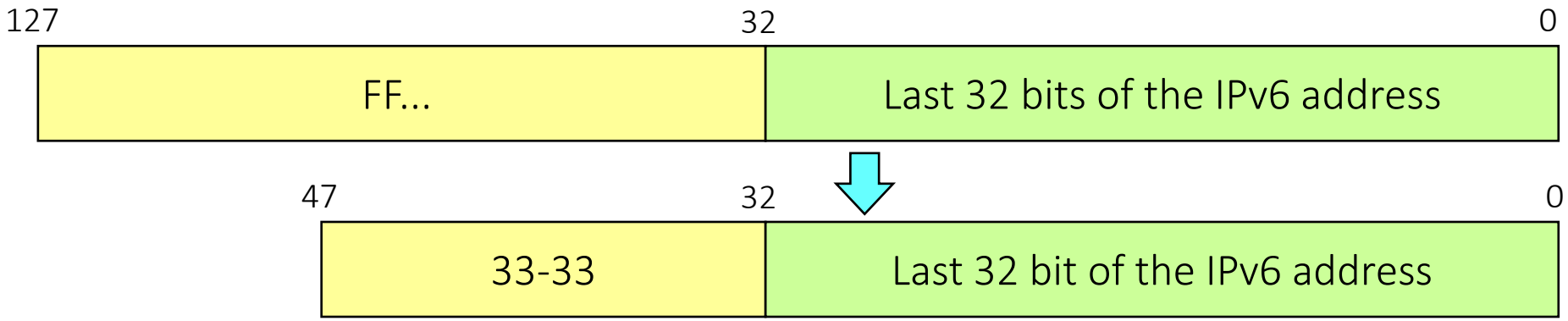
- IP unicast address
  - Procedural (protocol-based) discovery
  - Neighbor Discovery (see later)

- IP multicast address
  - Algorithmic mapping

IPv6 Multicast transmission

- Based on Ethernet multicast
  - Unlike an Ethernet broadcast, an Ethernet multicast can be filtered by the NIC

- IPv6 multicast address mapped to MAC address
  - 33-33-xx-xx-xx-xx is the reserved Ethernet MAC address when carrying an IPv6 multicast packet (RFC 7042)
  - the lower 32 bits of the MAC address (xx-xx-xx-xx) are copied from the lower 32 bits of the IPv6 multicast address



Example

- When sending a packet to the IP multicast address FFOC::89:AABB:CCDD
- The packet is encapsulated in a MAC frame with address 33:33:AA:BB:CC:DD

Neighbor Discovery and Address Resolution

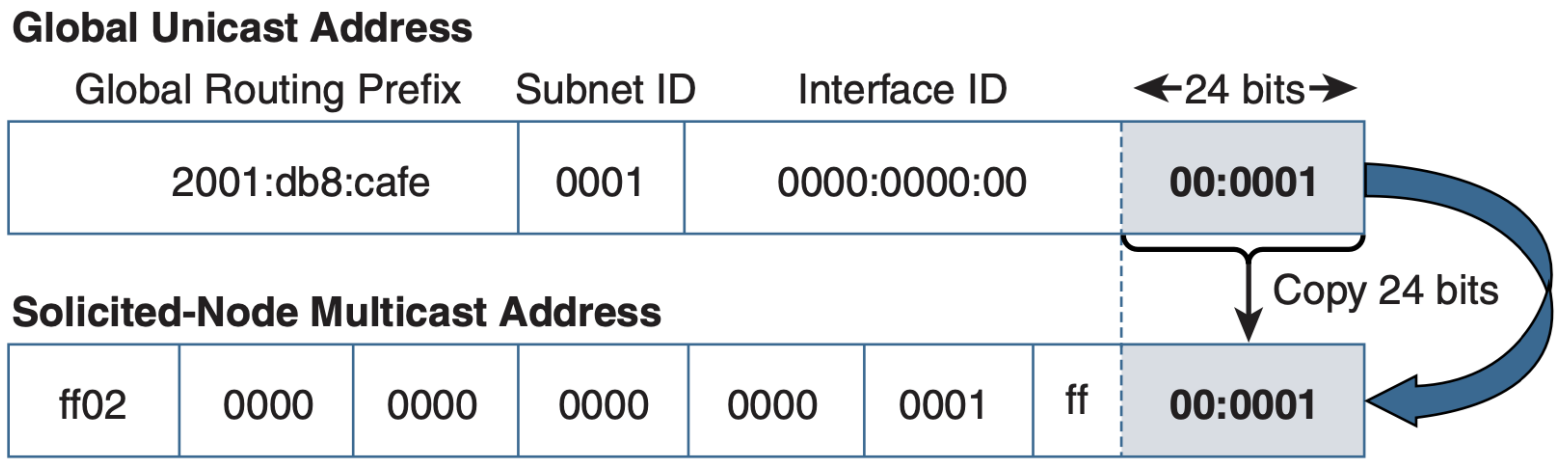


New function in ICMPv6

- It substitutes ARP
  - Based on multicast
    - Leverages the Solicited-node Multicast Address (FF02:0:0:0:0:1:FF00::/104)
  - Given the way multicast solicited address are constructed, most likely only one station gets involved

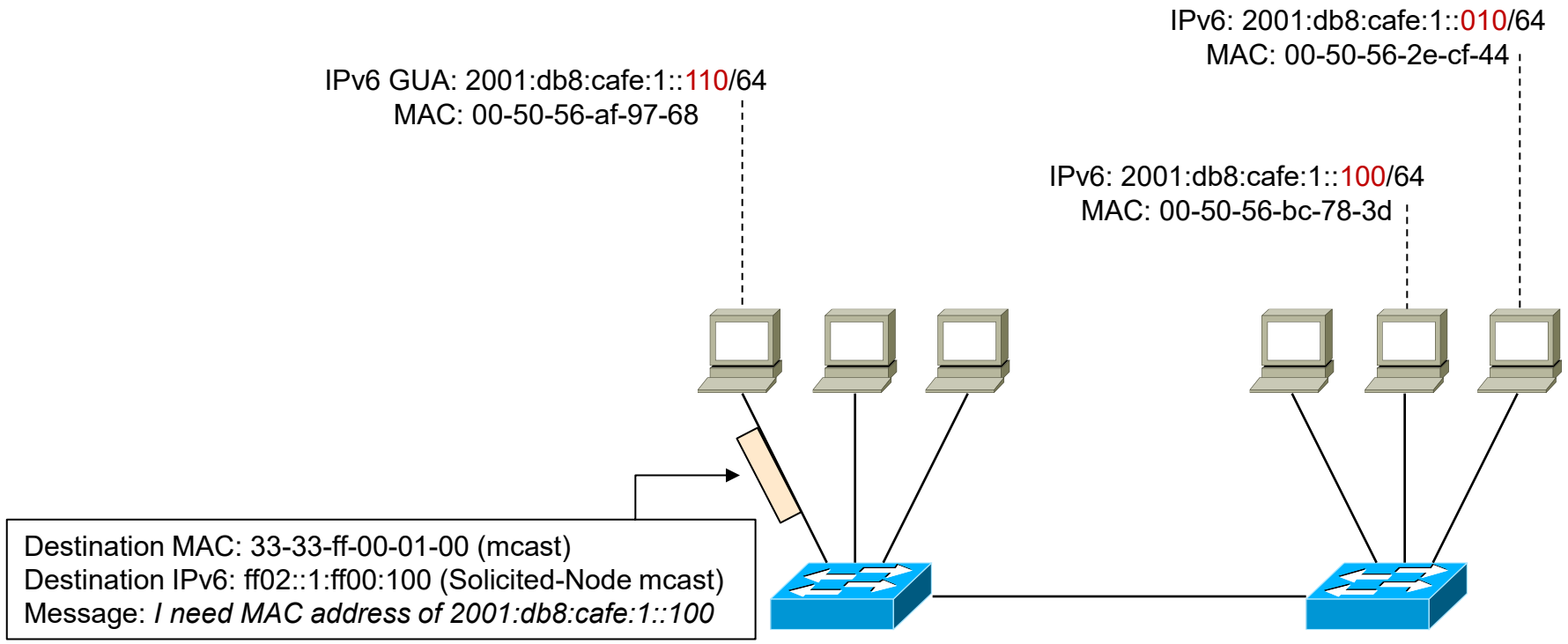
Solicited Node Multicast Address

- address automatically created for *every unicast address* on the interface.
- subscribed to by all hosts
- mapped as
  - FF02::1:FF/104 | 24 least significant bits of IP address
  - Likely one host per group



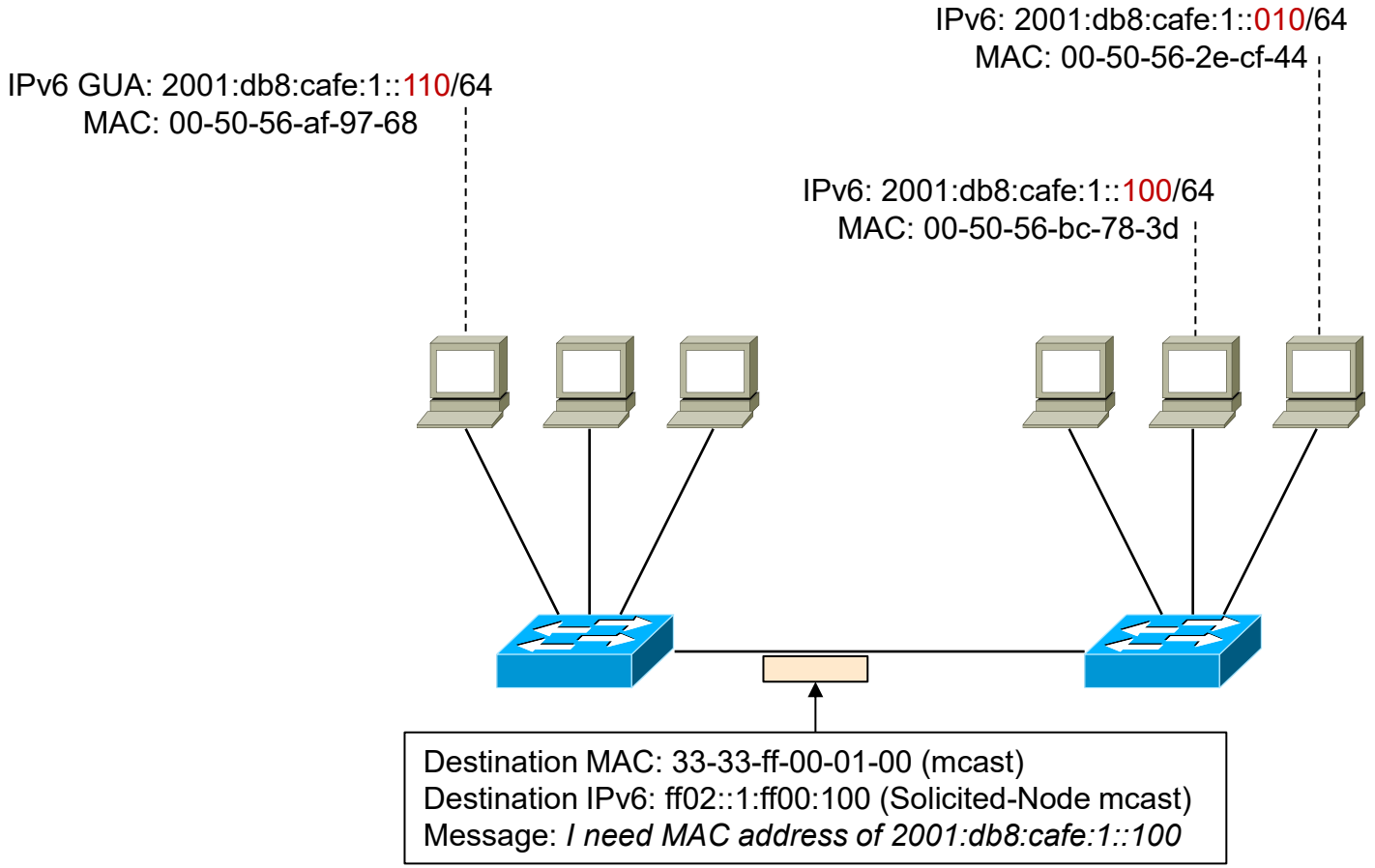
Address Resolution

- ICMP Neighbor Solicitation
  - Requester sends frame to Solicited Node Multicast Address of target IPv6 address



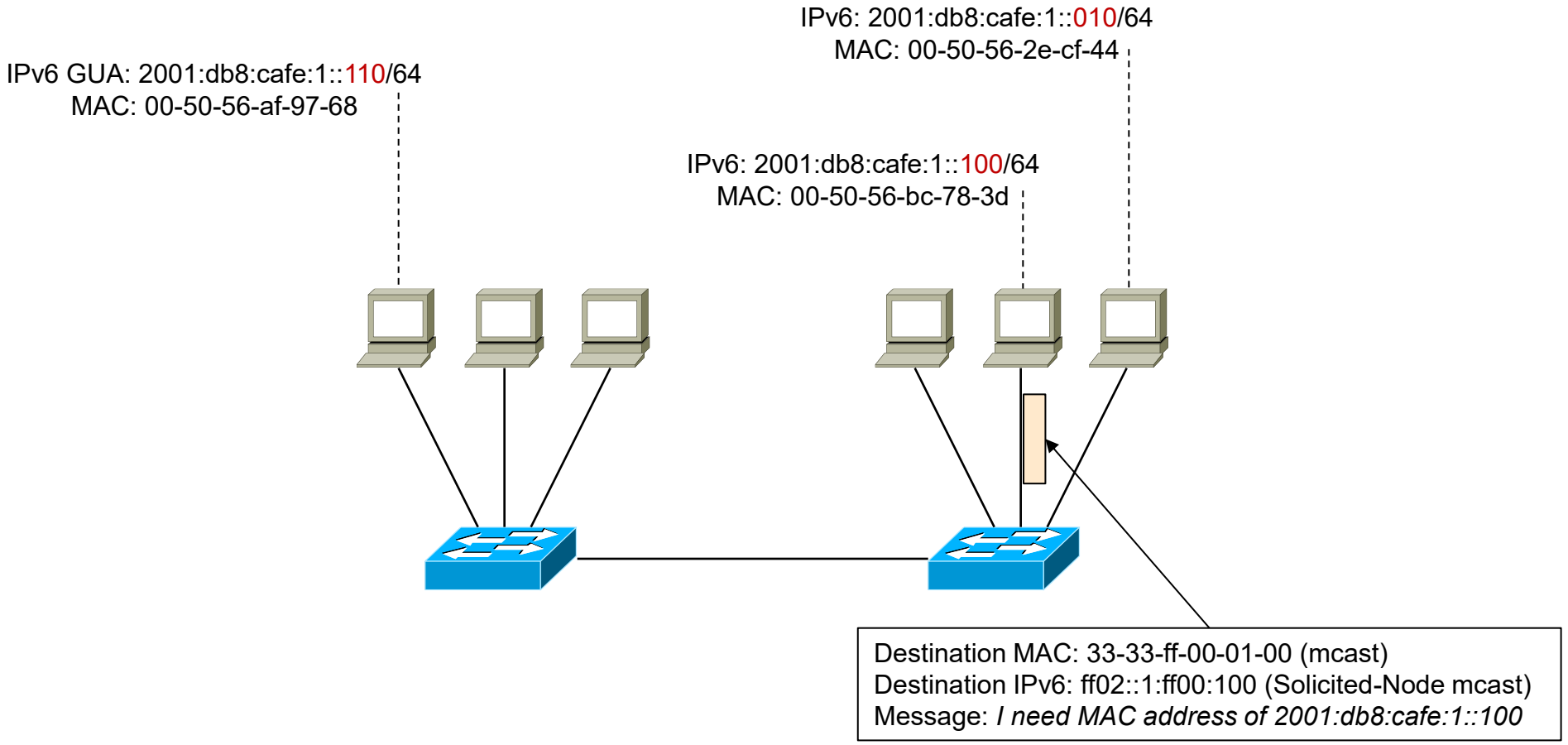
Address Resolution

- ICMP Neighbor Solicitation
  - Requester sends frame to Solicited Node Multicast Address of target IPv6 address



Address Resolution

- ICMP Neighbor Solicitation
  - Requester sends frame to Solicited Node Multicast Address of target IPv6 address



Address Resolution

- ICMP Neighbor Solicitation
  - Requester sends frame to Solicited Node Multicast Address of target IPv6 address
- ICMP Neighbor Advertisement
  - Sent back to requester's unicast address

- Mapping between IPv6 and MAC address stored in host cache
  - Equivalent to ARP cache

ICMPv6



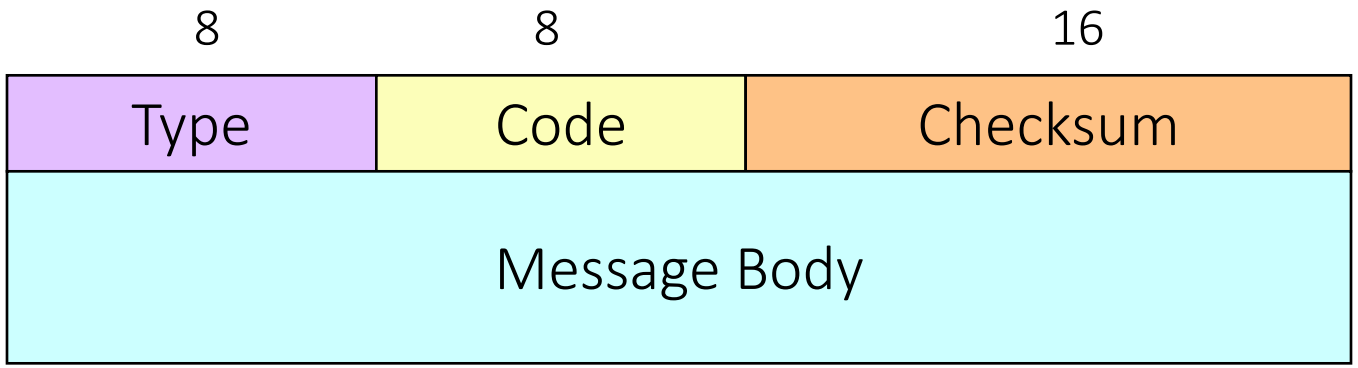
What is it used for?

- Diagnostics
- Neighbor Discovery
- Multicast group management
- Issue notification

- Includes functions that in IPv4 were delegated to
  - ARP (Address Resolution Protocol)
  - IGMP (Internet Group Membership Protocol)

Message Format

- Encapsulated in IPv6 packets
  - Next Header = 58
- Entire message length: max 65,495 bytes



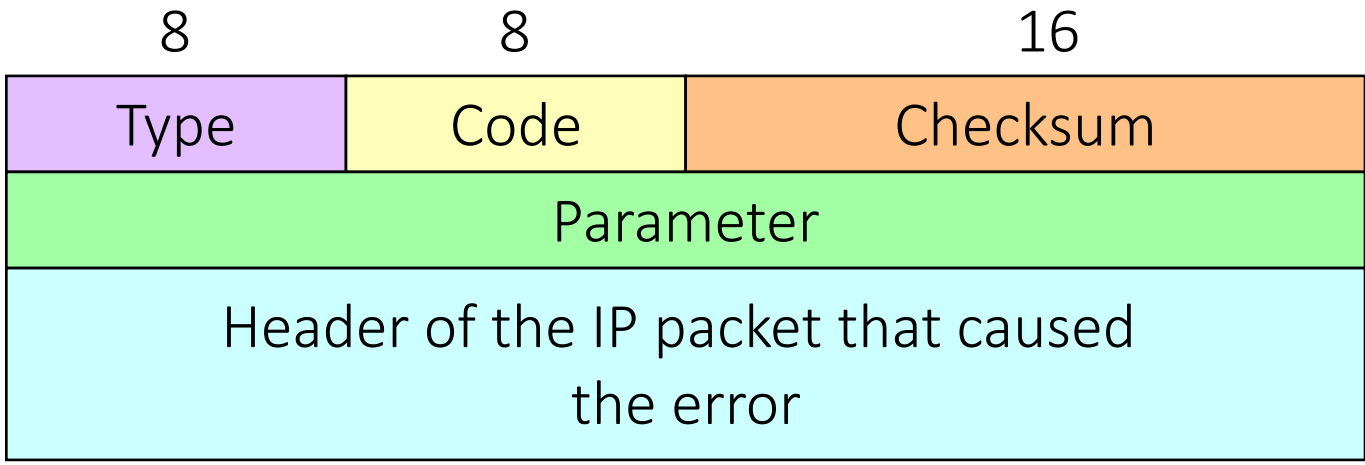
Type Field

Error messages (<128)

Informational messages (>=128)

1	Destination Unreachable
2	Packet too big
3	Time exceeded
4	Parameter Problem
128	Echo Request
129	Echo Reply
130	Multicast Listener Query
131	Multicast Listener Report
132	Multicast Listener Done
133	Router Solicitation
134	Router Advertisement
135	Neighbor Solicitation
136	Neighbor Advertisement
137	Redirect

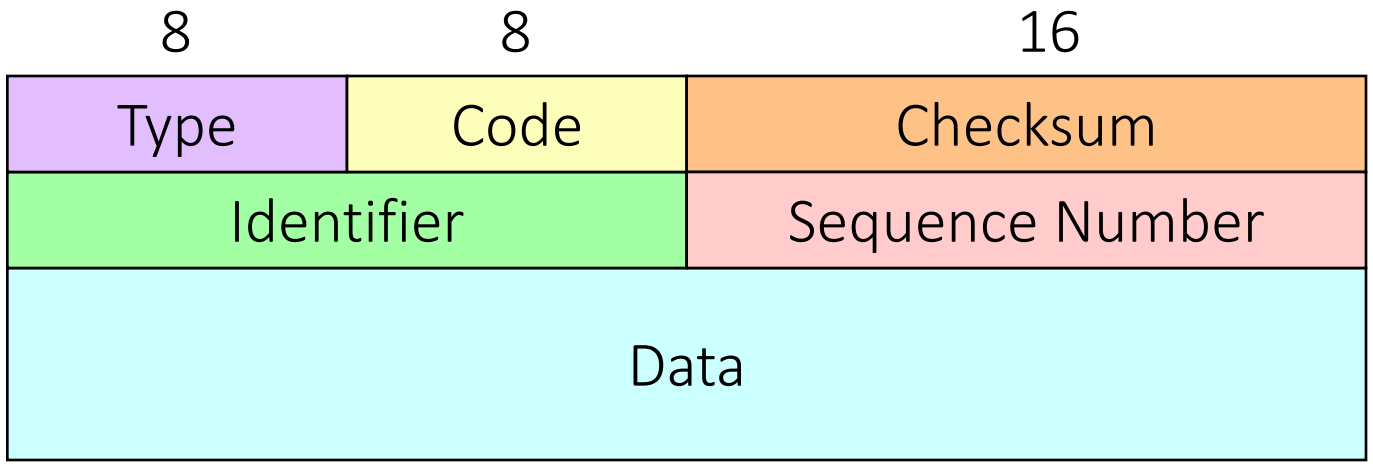
Error Messages



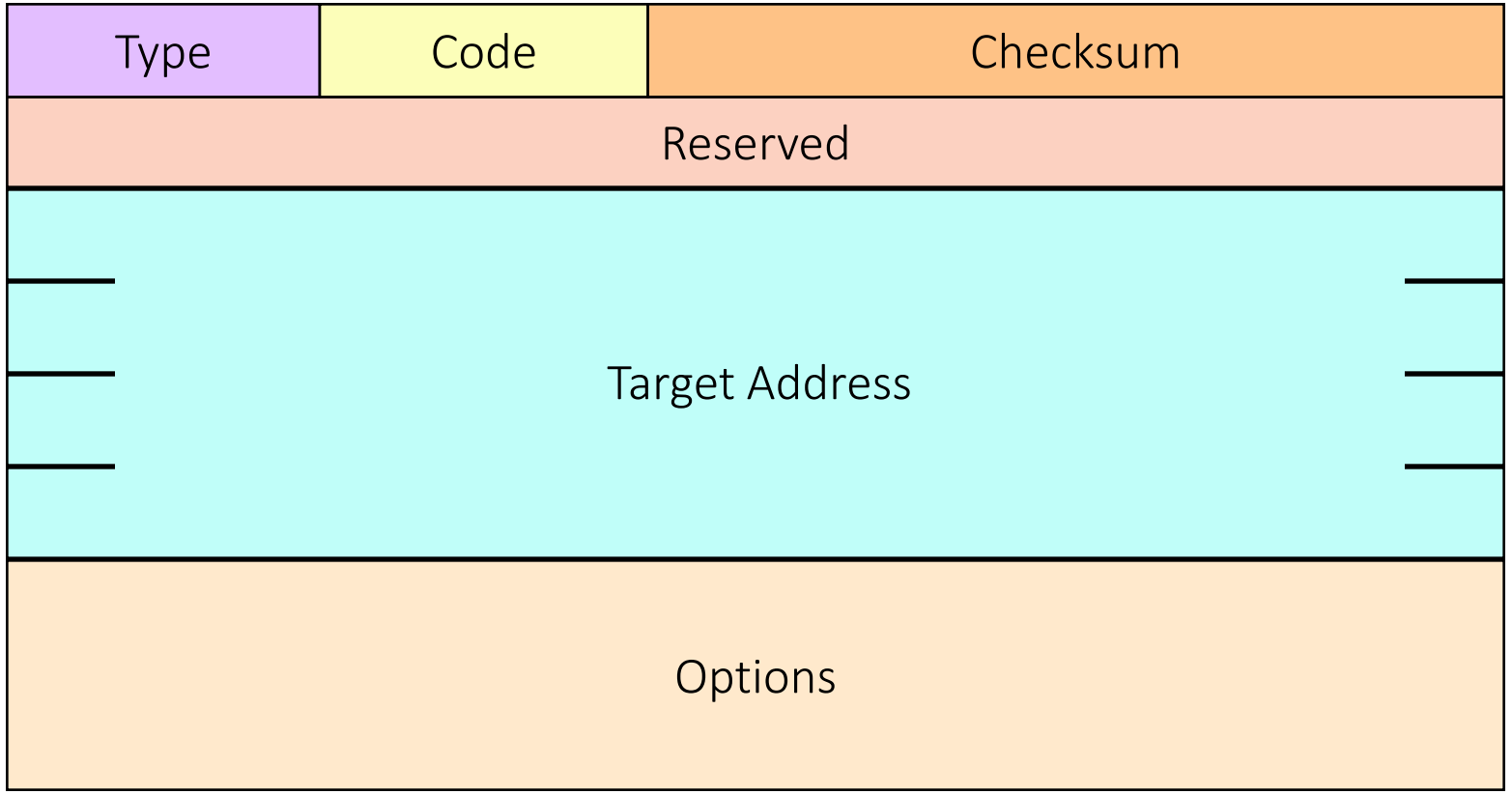
- Destination Unreachable (type = 1)
  - Usually generated by router or firewall
  - Code field provides reason (no route, wrong scope, address/port unreachable)
- Packet too big (type = 2)
  - IPv6 routers no longer fragment packets!
- Time exceeded (type = 3)
  - If router receives a packet with Hop Limit = 0
- Parameter Problem (type = 4)
  - generated when a device finds a problem with a field in the main IPv6 header or an extension header
  - example: invalid value in the Next Header field.

Informational Messages: Echo

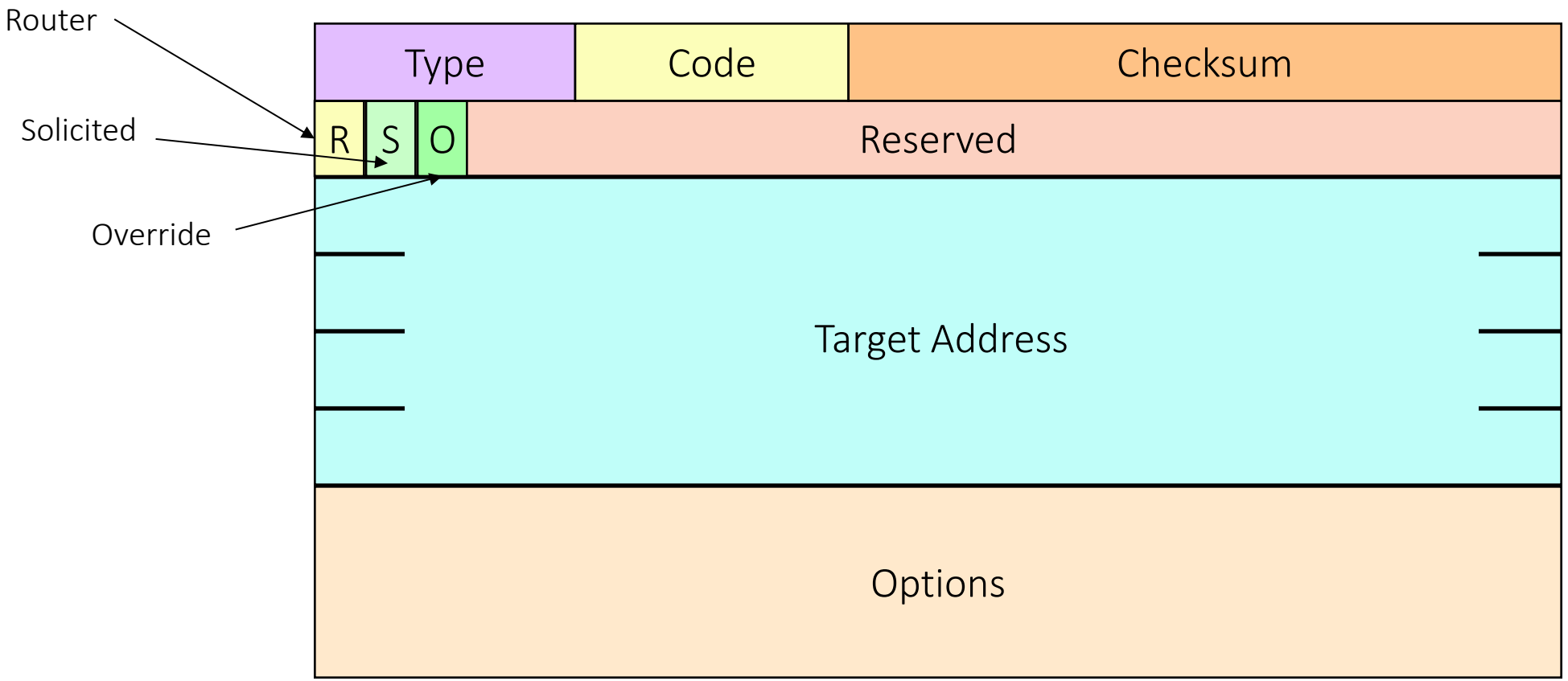
- Echo request (type= 128)
- Echo reply (type= 129)
- Used by `ping`



Informational Messages: Neighbor Solicitation



Informational Messages: Neighbor Advertisement

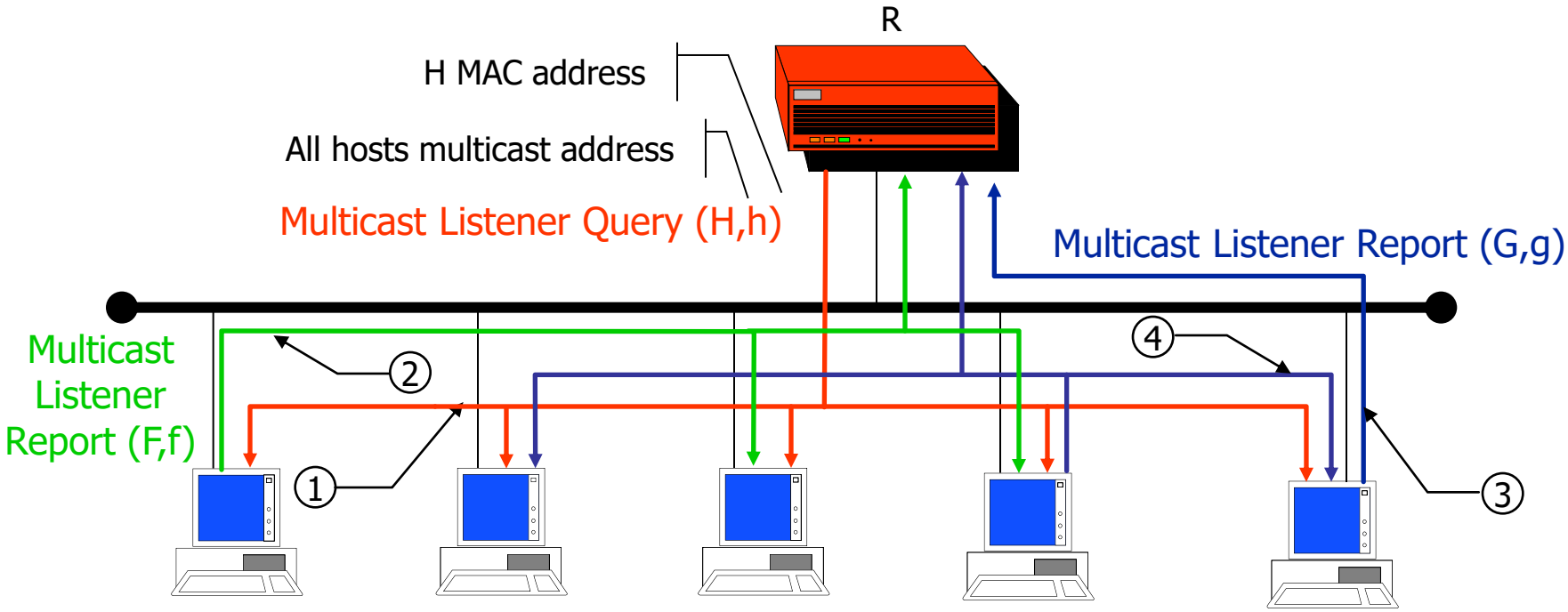


Multicast Group Management using ICMPv6

- Within a link, rely on data link layer multicasting service
  - Mapping of IPv6 multicast addresses onto MAC multicast addresses
- Among links, packets routed by routers
  - ICMPv6 to know on-link members
    - Hosts interests in receiving packets
  - Multicast routing protocols to know where there are off-link members

1	Destination Unreachable
2	Packet too big
3	Time exceeded
4	Parameter Problem
128	Echo Request
129	Echo Reply
130	Multicast Listener Query
131	Multicast Listener Report
132	Multicast Listener Done
133	Router Solicitation
134	Router Advertisement
135	Neighbor Solicitation
136	Neighbor Advertisement
137	Redirect

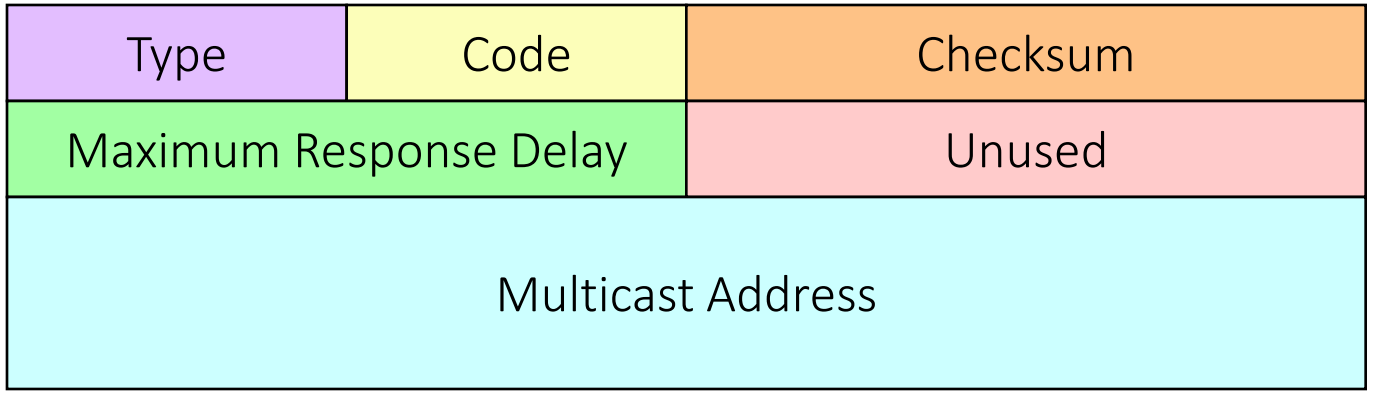
Host Membership Discovery



- R to announce multicast groups G and F
- R to forward packets to G and F

Informational Messages: Group Management

- Multicast Listener Query (type=130), sent by router to all-nodes-on-link multicast address (ff02::1)
  - General
  - Multicast group (address) specific
- Multicast Listener Report (type=131), sent by Listener to all-router multicast address (ff02::2)
- Multicast Listener Done (type=132), sent by Listener to all-router multicast address



Device Configuration in IPv6



What information is needed?

- Address prefix
- Interface identifier
- Default gateway
- DNS server
- Hostname
- Domain name
- MTU (Maximum Transmission Unit)
- …

Options

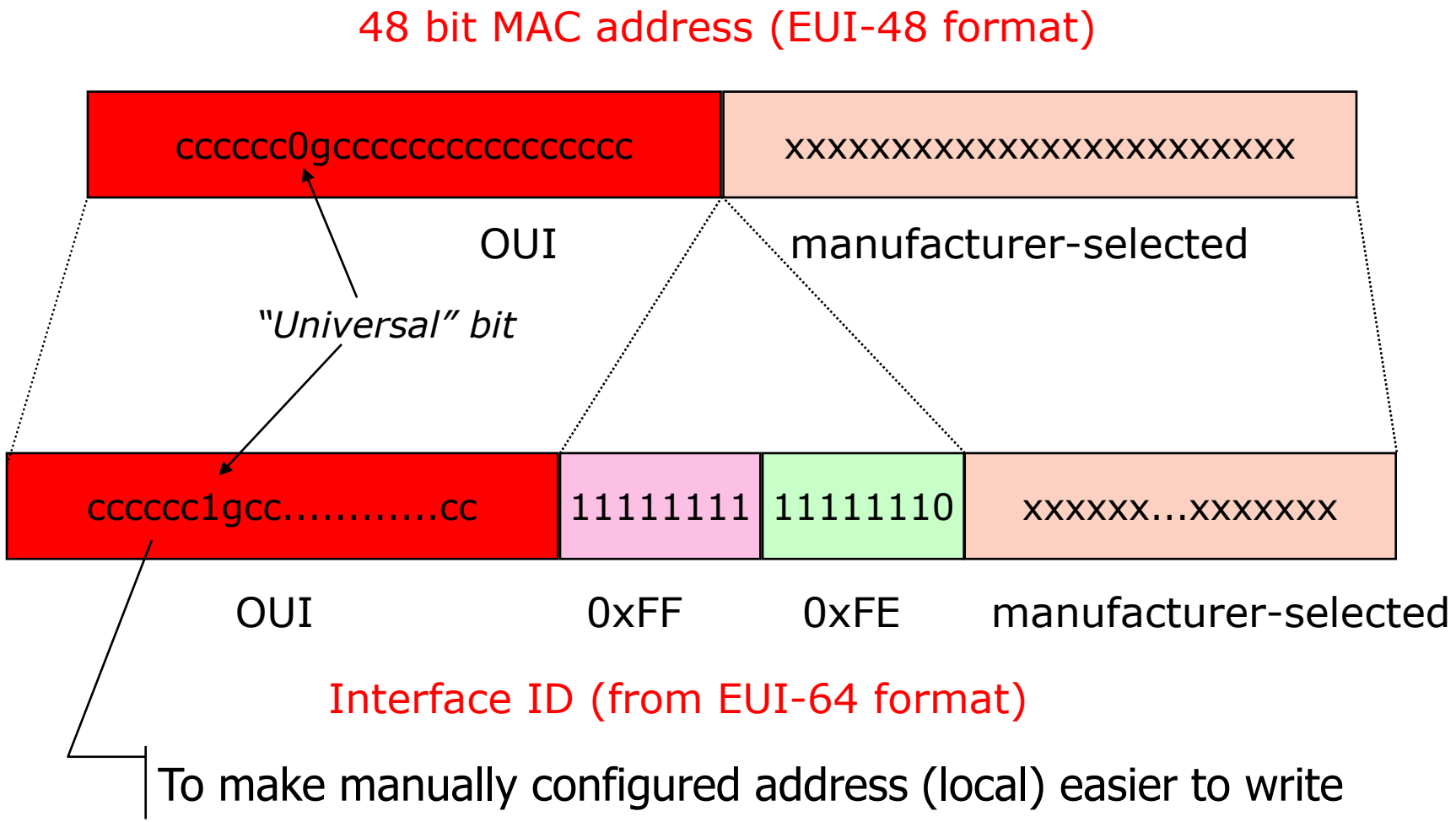
- Manual configuration
- Stateful configuration
  - Addresses and other parameters obtained through DHCP (DHCP6); the default gateway always comes from Router Advertisements (RA)
- Stateless configuration
  - Autogenerated
  - Address prefix obtained from RA
  - DNS servers can also come from the RA (RDNSS option, RFC 8106)
- Hybrid (Stateless DHCP)
  - Information other than address obtained through DHCP

Interface Identifier

- Manually configured
- Obtained through DHCPv6
- Automatically generated
  - From EUI-64 MAC address
  - Privacy aware

EUI-48 to EUI-64 mapping

EUI = Extended Unique Identifier

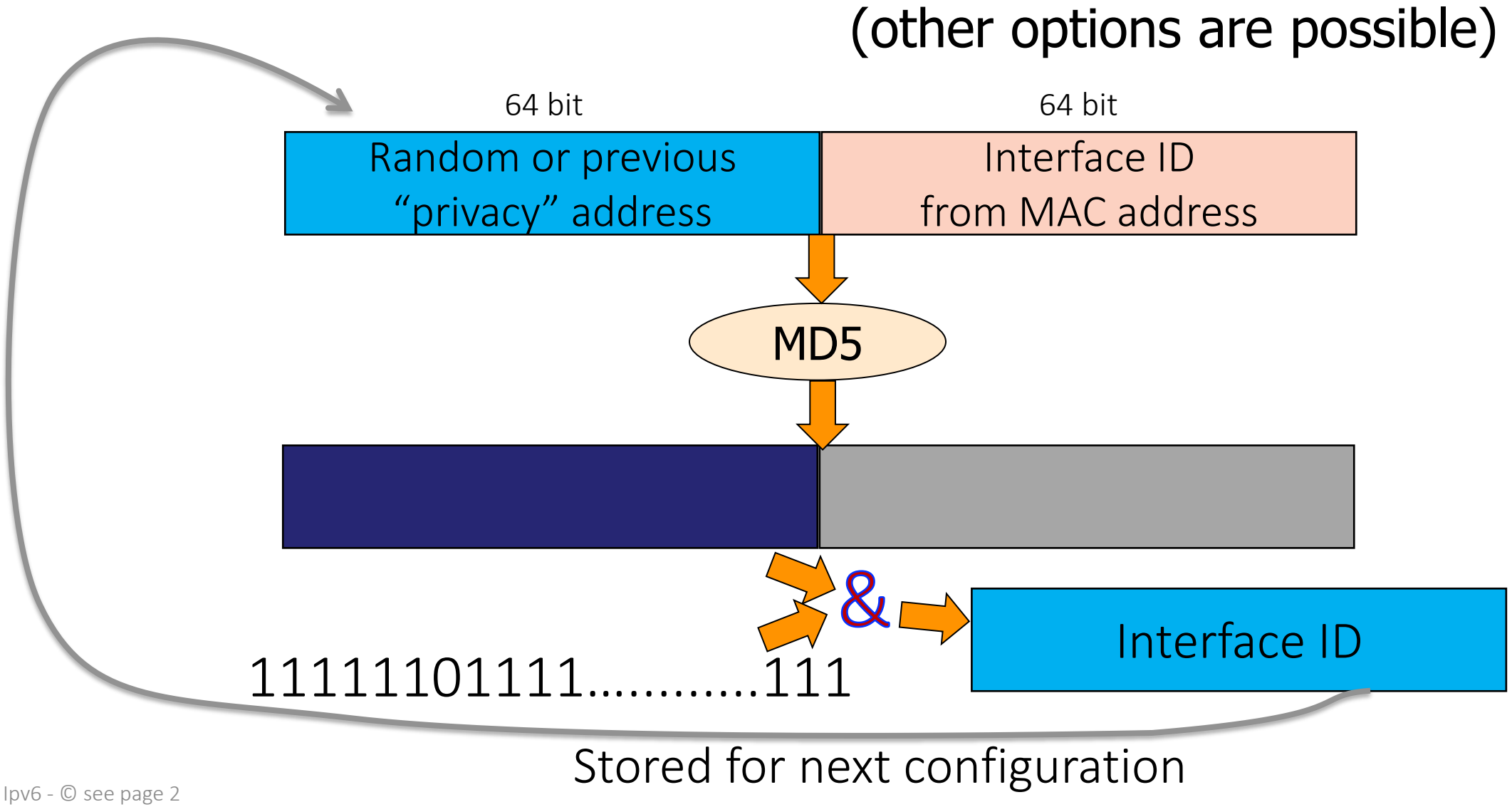


Privacy Concerns

- Traceability
  - The least significant 64 bits of the IPv6 address of an interface never change when MAC address is used
- RFC 4941, “Privacy Extensions for Stateless Address Autoconfiguration in IPv6”



Privacy Extension Algorithm



Address Usage

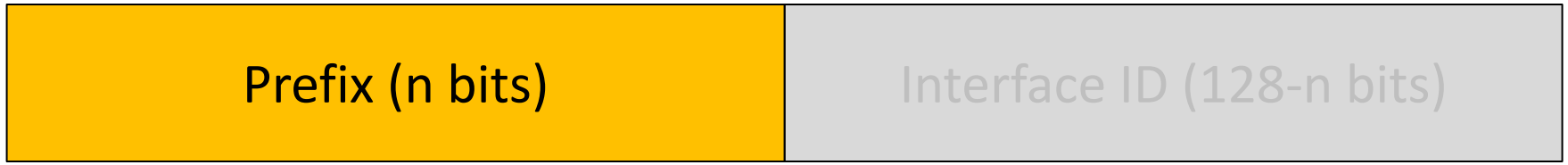
- A host may have several different addresses
  - “default”
  - “privacy-aware”

- Usable to accept/initiate connections
- Selection of address may be available to the user/application



Address Prefix Configuration

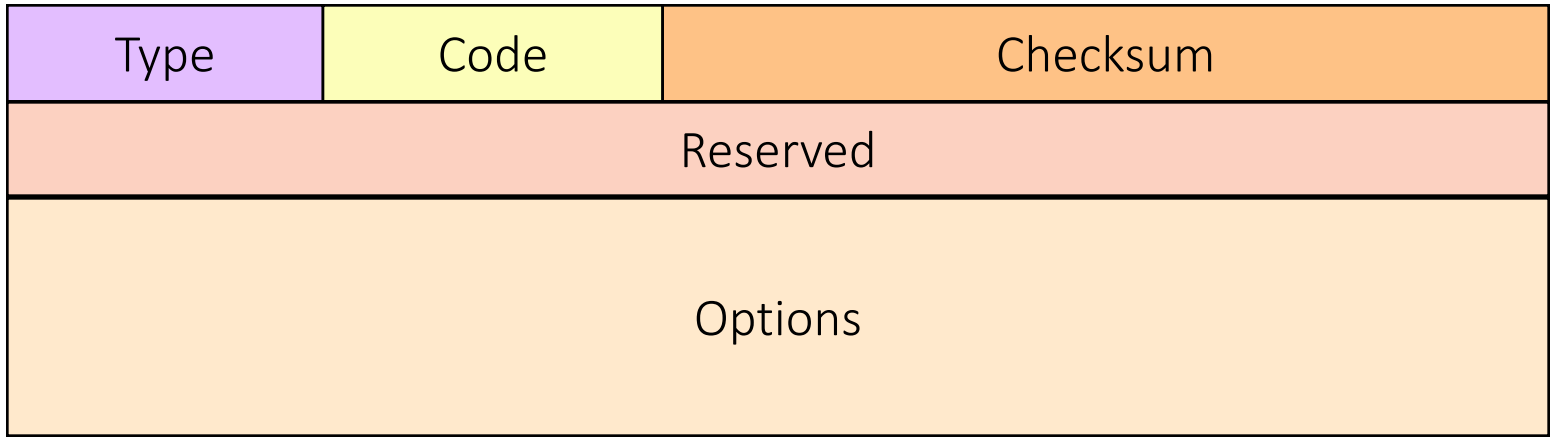
- Manually configured
- Obtained from DHCPv6
- Automatically generated
  - Link local
- Obtained from a router



Router/Prefix Discovery

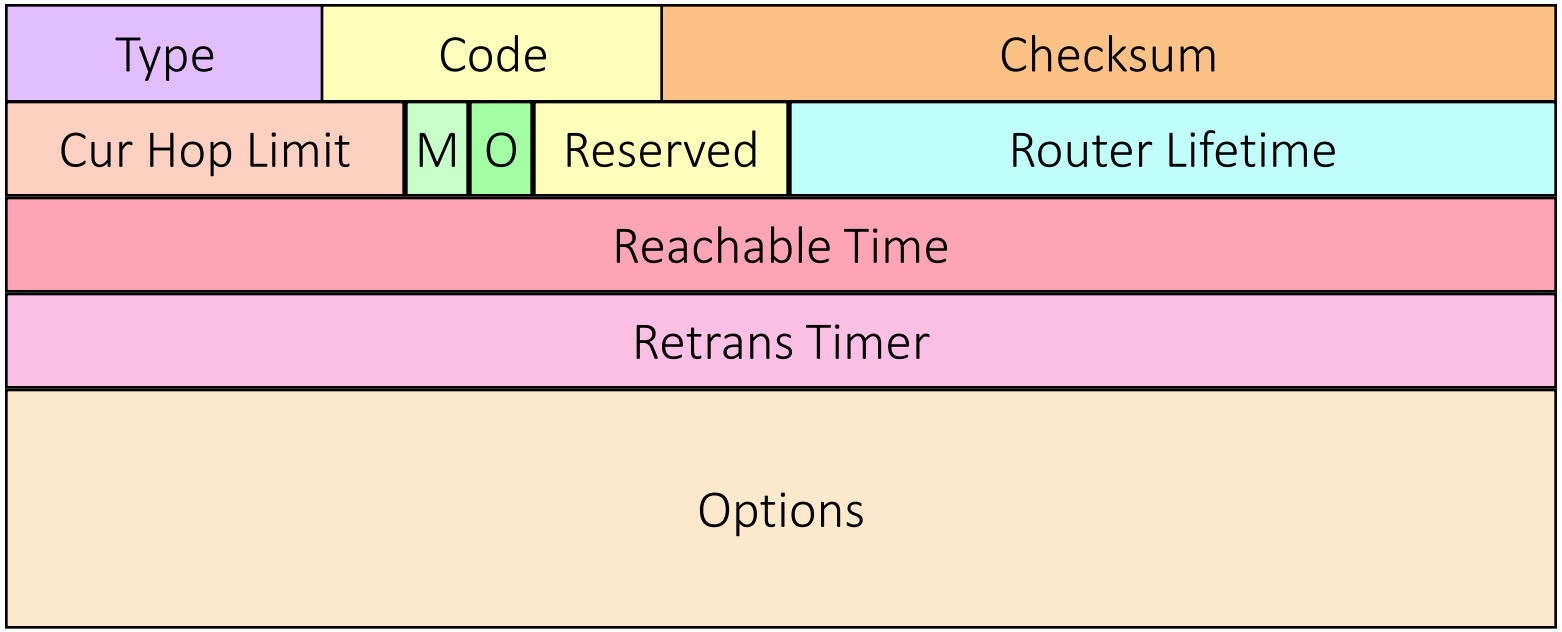
- ICMP Router Advertisement message
  - Sent by routers
- Solicited
  - Answering to Router Solicitation by host
- Unsolicited: periodic

Router Solicitation



Sent to the all-routers multicast address (FF02::2)

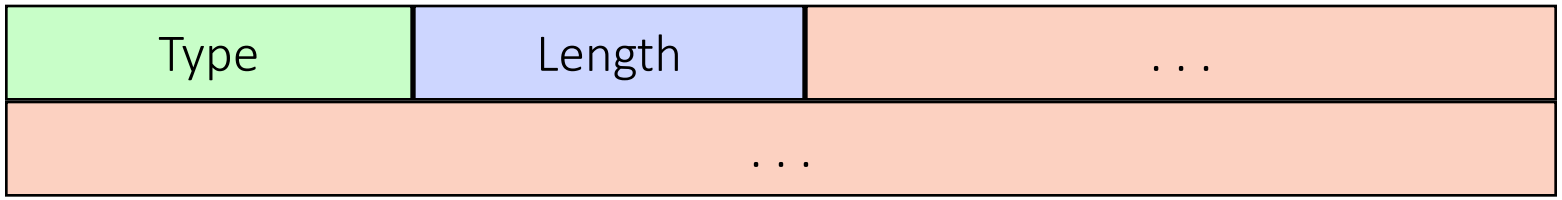
Router Advertisement



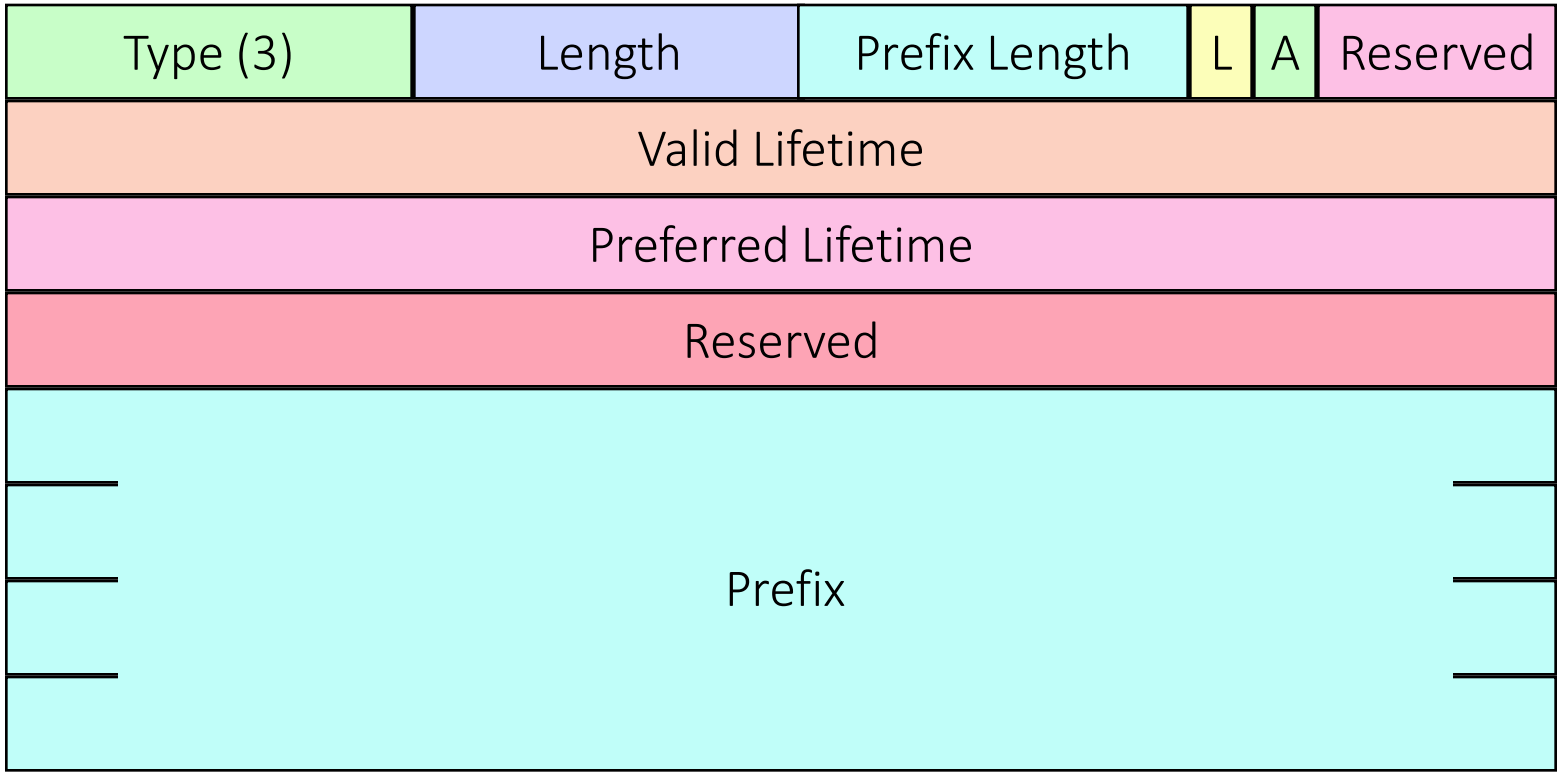
- M (Managed Address Configuration)
  - 1 – address available through DHCP
- O (Other configuration)
  - E.g., DNS server

Options

- General Format
- Length in multiple of 8 bytes



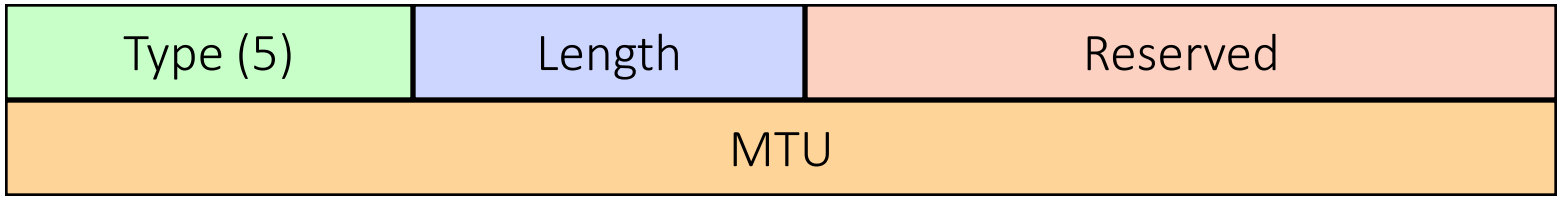
Prefix Information Option



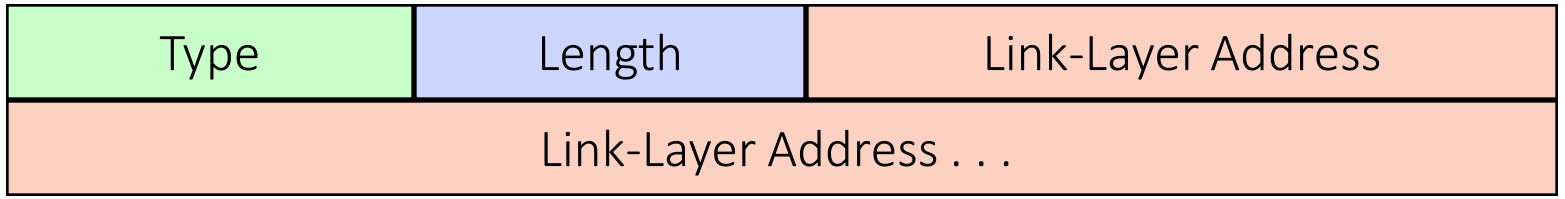
- L – prefix is on-link
- A – prefix can be used for autonomous configuration

MTU Option

- Ensures all hosts on-link use the same MTU value



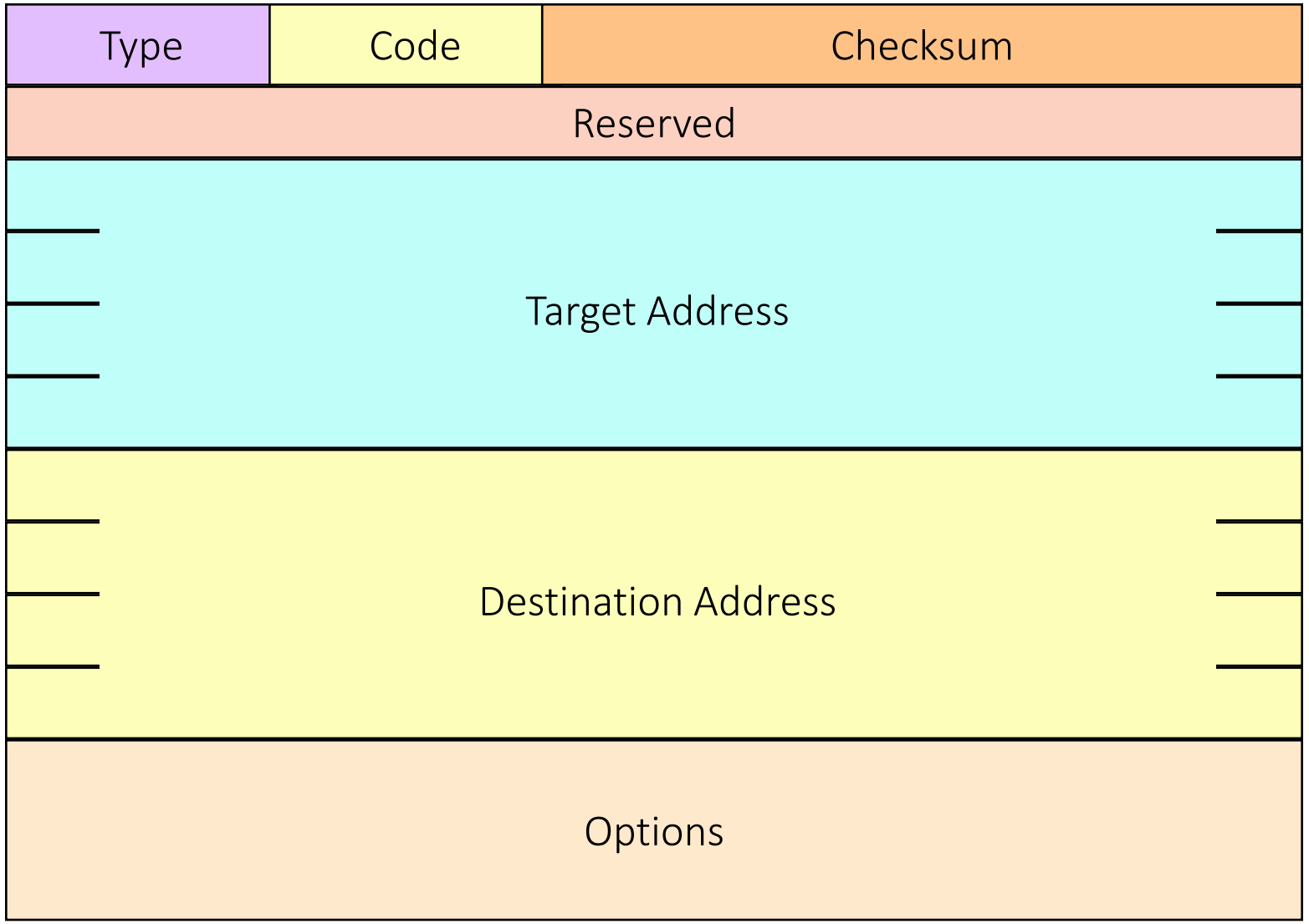
Link Layer Address Option



ICMP Redirect

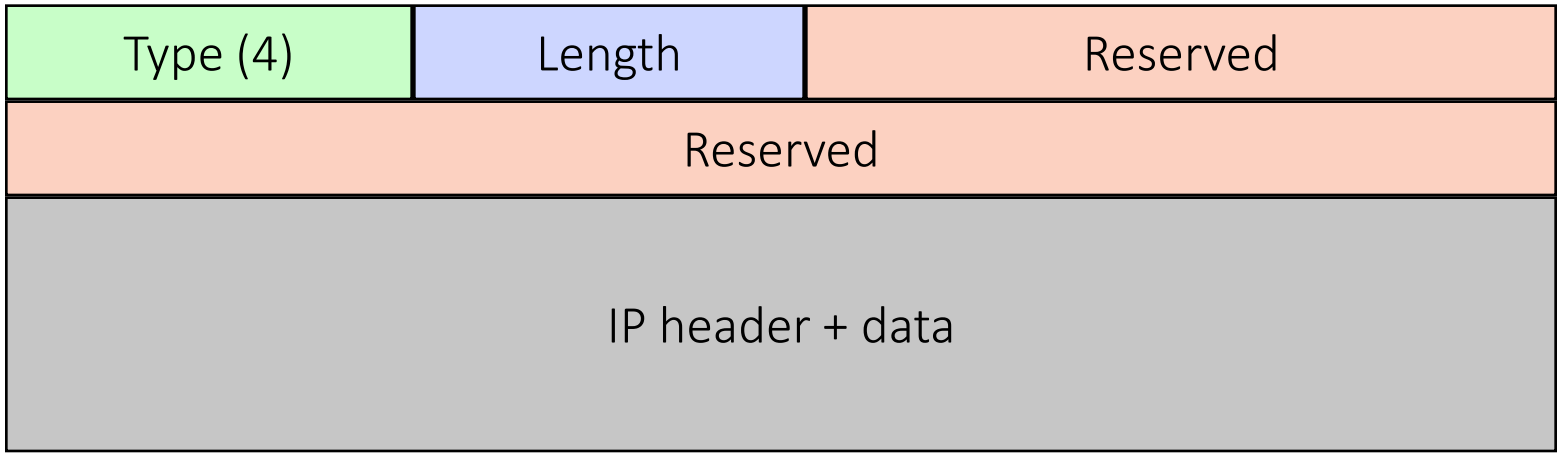
- Sent by a router to advise a host about a best first-hop
- The first-hop is always on-link, irrespective of prefix

ICMP Redirect Message Format



Redirect Header Option

Information about the packet being redirected



Duplicate Address Detection (DAD)

- Probe uniqueness of a newly-created IPv6 address
- Neighbor solicitation (NS) with address being probed as target
  - Source IPv6 address set to the unspecified address (::)
  - Sent to corresponding IPv6 Solicited Node Multicast Address
  - Incapsulated in a frame with corresponding MAC multicast address
- Wait for a response for at least 1 sec
  - If no Neighbor Advertisement is received, the new address is considered valid
  - If a Neighbor Advertisement is received, the new address cannot be used
- Before probing, the node joins the solicited-node group (MLD Report); an NS from another node probing the same address also means “duplicate”

Stateless Configuration: Basic Step

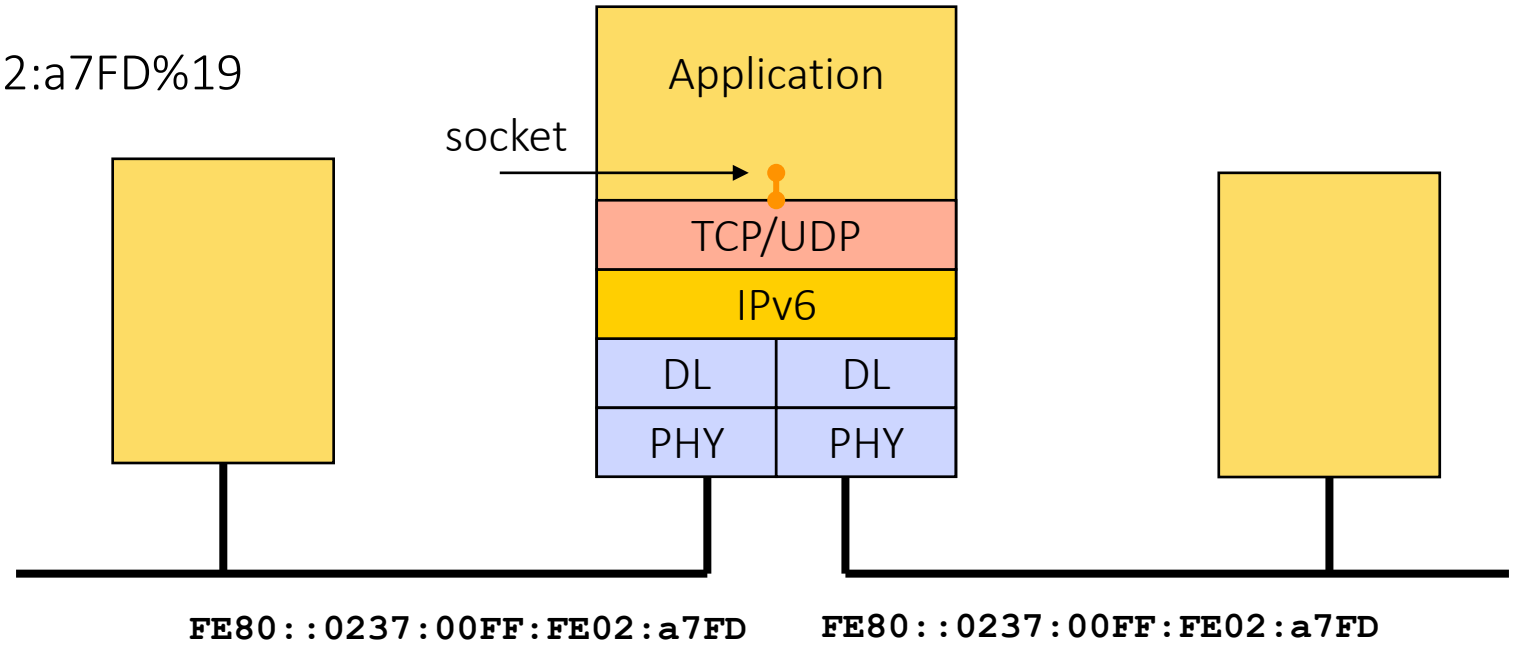
- Generate a link local address
- Probe for its uniqueness (DAD)
- Subscribe to the corresponding IPv6 Solicited Node Multicast Address
  - Configure reception of corresponding multicast MAC
  - Send ICMP Multicast Listener Report
- On-link communication enabled

Stateless Configuration: With Router

- Possibly send Router Solicitation
- Listen to Router Advertisements
- Create address from advertised prefix
- Probe for its uniqueness (DAD)
- Subscribe to the corresponding IPv6 Solicited Node Multicast Address
  - Configure reception of corresponding multicast MAC
  - Send ICMP Multicast Listener Report

Scoped addresses

- A scoped address is composed of an IPv6 address followed by a % and a number identifying the interface
- Used by the operating system to discriminate among IPv6 addresses on different interfaces with the same prefix (and possibly with the same interface ID)
- Example:
  - FE80::0237:00FF:FE02:a7FD%19



Examples of Scoped Addresses

```
c:\>netsh interface ipv6 show address

Interface 1: Loopback Pseudo-Interface 1

Addr Type  DAD State   Valid Life Pref. Life Address
---------  ----------- ---------- ---------- ------------------------
Other      Preferred     infinite   infinite ::1

Interface 10: Wireless Network Connection

Addr Type  DAD State   Valid Life Pref. Life Address
---------  ----------- ---------- ---------- ------------------------
Other      Preferred     infinite   infinite fe80::9832:45b1:96e9:f444%10

Interface 9: Local Area Connection

Addr Type  DAD State   Valid Life Pref. Life Address
---------  ----------- ---------- ---------- ------------------------
Other      Deprecated    infinite   infinite fe80::9158:6fc2:4155:3564%9

Interface 12: Local Area Connection* 12

Addr Type  DAD State   Valid Life Pref. Life Address
---------  ----------- ---------- ---------- ------------------------
Public     Preferred     infinite   infinite 2001:0:5ef5:79fd:14b0:f4d:f50d:a9a9
Other      Preferred     infinite   infinite fe80::14b0:f4d:f50d:a9a9%12

Interface 27: Bluetooth Network Connection

Addr Type  DAD State   Valid Life Pref. Life Address
---------  ----------- ---------- ---------- ------------------------
Other      Deprecated    infinite   infinite fe80::9961:aca4:ff3:3374%27

Interface 31: Local Area Connection* 25

Addr Type  DAD State   Valid Life Pref. Life Address
---------  ----------- ---------- ---------- ------------------------
Other      Deprecated    infinite   infinite fe80::5efe:10.242.86.86%31

c:\>
```